

# Explainable AI-Driven Remote Epilepsy Care Platform

DBA Deliverable — Real-Data EEG Pipeline, Governance, and Responsible AI

Patient index case: EP001 · Compiled from the project documentation set

# Contents

|    |                                                         |
|----|---------------------------------------------------------|
| 1  | 1. Overview                                             |
| 2  | 2. Global Policy                                        |
| 3  | 3. Enterprise Operating Model — 7 Pipelines / 40 Stages |
| 4  | 4. Pipeline 1 — Research & Clinical Protocol            |
| 5  | 5. Pipelines 2-7 + Every Missing Layer                  |
| 6  | 6. Prediction Output Schema                             |
| 7  | 7. Secondary (EEG) — 23-Phase Pipeline on Real Data     |
| 8  | 8. AIOps · DataOps · ModelOps · Fusion · Decision-AI    |
| 9  | 9. Vector DB Pipeline for RAG                           |
| 10 | 10. Security & Compliance (HIPAA/NIST/OWASP)            |
| 11 | 11. IRB Submission                                      |
| 12 | 12. Patient Consent + EULA                              |
| 13 | 13. Model Lifecycle — Phase-Gate Scorecard              |

# DBA Dissertation Blueprint

## Enterprise AI Platform for Explainable Multimodal Epilepsy Intelligence

**Why (this doc):** It is the single entry point that frames the entire dissertation — establishing the three central contributions, the five pillars, and how enterprise AI for **epilepsy** intelligence ties technical innovation to measurable business value.

**How:** By presenting a research-spine (problem → objective → contributions), a pillar-to-pipeline map, an end-to-end enterprise research stack, and diagrammatic overviews (flowchart, sequence, network, journey) so an examiner can grasp scope in minutes.

This repository holds the full dissertation blueprint and (subsequently) the reference implementation for a **Doctor of Business Administration (DBA)** focused on enterprise AI transformation in healthcare.

**Problem.** Epilepsy care is fragmented across clinical assessment and EEG interpretation, with limited explainable, governed AI support to help neurologists and EEG technicians reach consistent, defensible decisions at enterprise scale.

**Research Objective.** Design, build, and evaluate an explainable, multimodal (clinical + EEG) enterprise AI platform for epilepsy intelligence, and quantify its clinical, operational, and organizational impact — demonstrating DBA-level enterprise transformation rather than a purely technical artifact.

### The Three Central Contributions

**Why:** Focusing on three defensible contributions keeps the dissertation coherent and examinable rather than a scatter of disconnected features. **How:** Each contribution maps to a distinct layer — model framework, enterprise platform, and value evaluation.

Rather than presenting many independent innovations, the dissertation is organized around three central, defensible contributions:

1. **Multimodal AI Framework** — integrates primary clinical assessment with EEG analytics for epilepsy decision support.
2. **Enterprise Clinical Intelligence Platform** — combines explainable AI, evidence retrieval (RAG), human oversight, and governance.
3. **Business Value Evaluation Framework** — demonstrates how enterprise AI affects clinical workflows, operational efficiency, governance, and organizational outcomes.

These combine technical innovation with organizational transformation and measurable business impact — the defining posture of a DBA rather than a purely technical PhD.

*Caption - The flowchart below shows how the three contributions stack into one enterprise pipeline, from multimodal input to measured business value.*

■ Diagram (Mermaid — interactive in the viewer): flowchart TD

### The Five Pillars / Five Pipelines

**Why:** The five pillars decompose the platform into buildable, testable pipelines that each yield a concrete output. **How:** A pillar-to-pipeline table pairs purpose with final output so scope and deliverables are unambiguous.

Caption - This table anchors the whole blueprint: it enumerates each pillar, its pipeline, purpose, and final output, making the architecture and deliverables auditable at a glance.

| Pillar | Pipeline                                     | Purpose                                                               | Final Output                     |
|--------|----------------------------------------------|-----------------------------------------------------------------------|----------------------------------|
| 1      | <b>A – Primary Assessment AI</b> (16 phases) | Clinical assessment, questionnaires, medication, neurologist workflow | Clinical risk prediction         |
| 2      | <b>B – Secondary EEG AI</b> (16 phases)      | EEG signal processing, biomarkers, deep learning                      | EEG abnormality prediction       |
| 3      | <b>C – Multimodal Fusion AI</b>              | Combine clinical + EEG + longitudinal monitoring                      | Integrated patient risk score    |
| 4      | <b>D – Enterprise AI Platform</b>            | RAG, multi-agent orchestration, deployment, monitoring, governance    | Production hospital AI ecosystem |
| 5      | <b>E – Enterprise Evaluation</b>             | Clinical + AI + business + operations + governance validation         | Measurable organizational impact |

For the DBA, **Pipeline C (Multimodal Fusion)** is the primary research contribution — it enables the core experimental comparison: primary-only vs. EEG-only vs. multimodal.

Caption - The sequence diagram traces a single epilepsy case (EP001) through the roles and pipelines, clarifying how a Neurologist and EEG Technician interact with the platform in order.

■ Diagram (Mermaid — interactive in the viewer): sequenceDiagram

End-to-End Enterprise Research Stack

**Why:** The stack shows the linear research logic from a business problem to defensible DBA contributions, proving the work is driven by inquiry, not tooling. **How:** An ordered pipeline of stages, mirrored as a left-to-right network diagram for the enterprise view.

Caption - The network graph below renders the same enterprise stack as interconnected components, highlighting how data and evidence flow laterally across the platform rather than as an isolated chain.

■ Diagram (Mermaid — interactive in the viewer): graph LR

Business Problem

→ Research Questions

→ Literature Review

→ Primary Clinical AI

→ Secondary EEG AI

→ Multimodal Fusion AI

→ Explainable AI

→ Clinical Evidence RAG

→ Multi-Agent Enterprise AI

→ Clinical Decision Support

→ Enterprise Deployment

→ AI Governance

→ Business Evaluation

→ DBA Contributions

## Document Map

**Why:** The map orients readers to where each part of the argument lives across the eight-part dissertation. **How:** A part-by-part index links chapters to pipelines and appendices.

Caption - The user journey below models how an examiner or clinician moves through the document parts, capturing where confidence is built or where friction may occur during a defense read-through.

■ Diagram (Mermaid — interactive in the viewer): journey

- **Part I** — Business Problem (Chapters 1–3)
- **Part II** — Research Methodology (Chapter 4)
- **Part III** — Technical Framework (Chapters 5–9 / Pipelines A–E)
- **Part IV** — Implementation (Chapters 10–12)
- **Part V** — Evaluation (Chapters 13–17)
- **Part VI** — Results
- **Part VII** — Discussion
- **Part VIII** — Conclusion
- **Appendix** — Instruments, diagrams, model cards, checklists
- **Publication Strategy** — Six-paper Q1 output plan

## Professor Readiness (Defense Q&A)

**Why:** Anticipating examiner questions demonstrates command of the work's scope, novelty, and limits. **How:** Each likely question is posed as a heading with a concise, defensible answer.

### **Why is this a DBA and not a technical PhD?**

The central deliverable is not a single algorithm but an enterprise transformation: the platform couples multimodal epilepsy models with governance, explainability, human oversight, and a business value evaluation framework. Contributions 2 and 3 explicitly measure organizational, operational, and clinical-workflow impact, which is the defining posture of a DBA.

### **What is the core experimental comparison?**

The primary contribution, Pipeline C (Multimodal Fusion), enables a controlled comparison of **primary-only vs. EEG-only vs. multimodal** epilepsy risk prediction. This isolates the incremental value of fusing clinical assessment with EEG analytics, using test patient EP001 and the wider cohort.

### **How do the two clinical roles interact with the platform?**

The **Neurologist** owns clinical assessment entry and the final confirm-or-override decision, while the **EEG Technician** captures and uploads EEG recordings for Pipeline B. The platform fuses both streams and returns an explainable recommendation, keeping a human in the loop for every decision.

### **How is trust and governance addressed?**

Through the Enterprise AI Platform (Pillar D): explainable AI surfaces the drivers of each prediction, RAG grounds recommendations in retrieved clinical evidence, and the governance layer enforces oversight, monitoring, and auditability — aligning with responsible-AI expectations for high-stakes clinical settings.

## References

**Why:** Grounding the blueprint in authoritative sources establishes its clinical and AI-governance credibility. **How:** APA 7th edition entries spanning epilepsy classification, medical AI, and responsible-AI standards.

- American Psychological Association. (2020). *Publication manual of the American Psychological Association* (7th ed.). American Psychological Association.
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# GLOBAL POLICY (Mandatory)

These rules are **mandatory** for every document and code change in this project. They apply automatically — no need to restate them per task.

| #  | Policy                                | Rule                                                                                                                                                                                       |
|----|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | <b>Scope</b>                          | Epilepsy only. Exclude schizophrenia/PANSS/psychiatry; translate any such structure to epilepsy (Neurologist + EEG Technician).                                                            |
| 2  | <b>Order</b>                          | Docs first, then code.                                                                                                                                                                     |
| 3  | <b>Format</b>                         | All data presented in Markdown tables.                                                                                                                                                     |
| 4  | <b>Granularity</b>                    | One MD file per unit of data (per assessment / phase / role).                                                                                                                              |
| 5  | <b>Diagrams</b>                       | Every doc must present: <b>flowchart</b> , <b>sequence diagram</b> , <b>network diagram</b> , <b>journey map</b> (Mermaid).                                                                |
| 6  | <b>Citations</b>                      | APA 7th edition formatting in every References section.                                                                                                                                    |
| 7  | <b>UX</b>                             | Advanced readability, clear flow, polished UI/UX in the viewer.                                                                                                                            |
| 8  | <b>Tracking</b>                       | Terminal task list kept current every turn (done vs pending).                                                                                                                              |
| 9  | <b>Prompt log</b>                     | Save each prompt input + response output to docs/prompt - log/.                                                                                                                            |
| 10 | <b>Tooling</b>                        | Install any needed software (pre-approved).                                                                                                                                                |
| 11 | <b>Top-1% bar</b>                     | Brutal self-review each pass: name what's missing for a top-1% DBA and close gaps. See GAP-ANALYSIS.md.                                                                                    |
| 12 | <b>References</b>                     | Every substantive doc ends with an APA7 <b>References</b> section.                                                                                                                         |
| 13 | <b>Explainability / justification</b> | Every heading and subheading carries a short <b>Why</b> (reason it exists) and <b>How</b> (method), so each level is self-explaining.                                                      |
| 14 | <b>Professor readiness</b>            | Every doc ends with a <b>Professor Readiness (Defense Q&amp;A)</b> section: hypothetical examiner questions as sub-headings, each answered concisely via step/table/flowchart/1–3 lines.   |
| 15 | <b>Research spine</b>                 | Every doc contains, in order: <b>Problem</b> → <b>Sub-problems</b> → <b>Research Problem</b> → <b>Research Objective</b> → <b>Flow</b> → <b>Hypotheses</b> → <b>Statistical Analysis</b> . |
| 16 | <b>Step duality</b>                   | Every step is explained with <b>both a table and a flowchart</b> .                                                                                                                         |
| 17 | <b>Table captions</b>                 | Every table is preceded by a 1–2 line note explaining <i>why that table is present</i> .                                                                                                   |
| 18 | <b>Progress cadence</b>               | Whenever a long-running background task (workflow/build) is active, post a status update to the user every ~5 minutes until it completes.                                                  |

| #  | Policy                                                                                               | Rule                                                                                                                                                                                                                                                                                                              |
|----|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 | <b>Standing approval</b>                                                                             | Do not pause to ask for approval. Proceed autonomously under this policy — install tooling, run workflows, write files, and complete every task. Only stop for genuinely destructive/irreversible actions.                                                                                                        |
| 20 | <b>Auto-resume after session limit</b>                                                               | If a workflow agent fails with a session/usage limit, schedule a wakeup for just after the stated reset time and automatically resume the workflow (resumeFromRunId, which replays cached successes and re-runs only the failed agents). Report which items were deferred.                                        |
| 21 | <b>C4 + detailed diagram explanation</b>                                                             | In addition to flowchart/sequence/network/journey, include a <b>C4-style model</b> (context/container/component) where architecture is discussed. Every diagram, model, matrix, and accuracy figure gets a detailed prose explanation: <b>Reason · Why · What is happening · How it is happening · Reference.</b> |
| 22 | <b>Code comments + docstrings</b>                                                                    | All code (scripts, generators, React components, workflow scripts) must carry docstrings/header comments and inline comments explaining intent.                                                                                                                                                                   |
| 23 | <b>README</b>                                                                                        | Maintain a detailed top-level README.md (purpose, structure, how to run, standards, data, index).                                                                                                                                                                                                                 |
| 24 | <b>Save all inputs</b>                                                                               | Every user input/prompt is recorded in docs/prompt-log/.                                                                                                                                                                                                                                                          |
| 25 | <b>Missing-check</b>                                                                                 | On each verification pass, actively check for anything missing (files, diagrams, references, sections) and record it in docs/COVERAGE-MATRIX.md with a fix.                                                                                                                                                       |
| 26 | <b>Security-grade</b>                                                                                | Every data path is security-grade by design: <b>HIPAA</b> Security/Privacy Rule, <b>NIST</b> CSF + 800-53 + AI RMF, <b>OWASP</b> Top-10 (web + LLM). See governance/00-security-compliance.md.                                                                                                                    |
| 27 | <b>Encryption</b>                                                                                    | PHI encrypted <b>at rest (AES-256)</b> and <b>in transit (TLS 1.2+)</b> ; keys in KMS (rotated); de-identify (Safe-Harbor) before analysis; no secrets/raw EEG in the repo.                                                                                                                                       |
| 28 | <b>Consent + EULA</b>                                                                                | Before any patient data is used for analysis, capture <b>informed consent + EULA</b> (versioned, timestamped, revocable). See governance/02-patient-consent-eula.md.                                                                                                                                              |
| 29 | <b>IRB</b>                                                                                           | Human-subjects data use requires an <b>IRB</b> protocol/approval on file. See governance/01-irb-submission.md.                                                                                                                                                                                                    |
| 30 | <b>Every phase = pipeline + analysis + monitoring + scoring + visualization + dashboard + report</b> | Each model/data phase must implement (or document) all seven: statistical + ML + subjective/sensitivity analysis, continuous monitoring (drift/concept-drift/alerts), scoring, a visualization, a dashboard, and a report.                                                                                        |



| #  | Policy                   | Rule                                                                                                                                                                                                                                                                                                                                         |
|----|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 | <b>Full analysis set</b> | Each dataset/model carries: EDA + class-balance (with SMOTE), feature engineering + selection, HPO, full <b>accuracy matrix</b> (acc/precision/recall/specificity/F1/AUC/AP/log-loss), loss curve, hypothesis test, sensitivity/ablation, SHAP/permutation explainability, before/after data visualization — all presented in the viewer UI. |

## Diagram Standard (Mermaid)

| Diagram     | Mermaid type    | Purpose                      |
|-------------|-----------------|------------------------------|
| Flowchart   | flowchart TD    | Process/decision flow        |
| Sequence    | sequenceDiagram | Actor interactions over time |
| Network     | graph LR        | Component/data relationships |
| Journey map | journey         | Patient/clinician experience |

## Test Patient

All worked examples use **EP001** (EP-2026-001) — see [primary-assessment/index.md](#).

# Enterprise Operating Model — Seven Connected Pipelines (Epilepsy)

**Why (this doc):** An expert review found the platform strong on analytics/modelling but weak on **system boundaries and operational ownership** — research methodology, data engineering, MLOps, GenAI, clinical operations, and governance were fused into one flow. This doc re-frames the whole platform as **seven connected pipelines**, each with its own owner, inputs, outputs, and controls, and maps the **40-stage enterprise architecture** onto them. **How:** each stage cites the module/doc that implements it or marks it a documented gap. **Scope: epilepsy only** (the reviewer's schizophrenia/PANSS/relapse examples are translated to epilepsy — see the translation key below).

## Translation key (schizophrenia → epilepsy)

*Caption — the reviewer used psychiatry examples; every one maps cleanly to an epilepsy analogue.*

| Reviewer term               | Epilepsy analogue used here                                           |
|-----------------------------|-----------------------------------------------------------------------|
| Schizophrenia diagnosis     | Epilepsy diagnosis / seizure-disorder classification (ILAE 2017)      |
| PANSS total / severity      | Seizure-severity level L1–L4 + seizure frequency + EEG severity       |
| Relapse within 90 days      | <b>Breakthrough seizure / seizure recurrence within 90 days</b>       |
| Psychiatric hospitalization | Seizure-related ED visit / status-epilepticus admission               |
| Antipsychotic / adherence   | Anti-seizure medication (ASM) / ASM adherence                         |
| Psychiatrist                | Neurologist / epileptologist                                          |
| Sleep, mood, functioning    | Sleep (a real seizure trigger), stress, ASM side-effects, functioning |

## The seven pipelines

*Caption — each pipeline is a bounded operating unit with a clear owner; arrows are hand-offs, not code coupling.*

| #  | Pipeline                                   | Owner                                          | Answers                                                                      | Key artefacts                                     |
|----|--------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------|
| P1 | <b>Research &amp; Clinical Protocol</b>    | DBA researcher + epileptologist + statistician | <i>What clinical problem, which population, what outcome?</i>                | protocol, cohort def, target def, SAP, ethics     |
| P2 | <b>Data Engineering &amp; Governance</b>   | Data architect + data steward                  | <i>How does data arrive, get contracted, governed, versioned?</i>            | data contracts, lakehouse zones, catalog, lineage |
| P3 | <b>Data Prep &amp; Feature Engineering</b> | ML/data engineer                               | <i>How is data cleaned, labelled, made leakage-safe, stored as features?</i> | feature registry, feature store, leakage review   |

| #  | Pipeline                                                | Owner                                  | Answers                                                             | Key artefacts                                      |
|----|---------------------------------------------------------|----------------------------------------|---------------------------------------------------------------------|----------------------------------------------------|
| P4 | <b>Statistical &amp; ML Modelling</b>                   | ML architect + statistician            | <i>What model, validated how, calibrated, fair?</i>                 | experiments, baselines, validation, calibration    |
| P5 | <b>MLOps &amp; Deployment</b>                           | ML platform engineer                   | <i>How is the model registered, served, rolled back, monitored?</i> | registry, serving, rollback, monitoring            |
| P6 | <b>RAG, LLM &amp; Agent Engineering</b>                 | GenAI engineer                         | <i>How is knowledge retrieved and generated safely?</i>             | vector DB, prompt registry, agent policy, LLM eval |
| P7 | <b>Clinical Safety, Responsible-AI &amp; Monitoring</b> | Clinical safety lead + privacy officer | <i>How is it kept safe, fair, private, and overseen?</i>            | safety layer, fairness, consent, audit             |

■ Diagram (Mermaid — interactive in the viewer): flowchart LR

## The 40-stage architecture, mapped to pipelines & status

Caption — the reviewer's 40-stage enterprise flow, each stage assigned to a pipeline and marked implemented / partial / documented-gap with the artefact that proves it.

| #  | Stage                                    | Pipeline | Status     | Evidence / gap                           |
|----|------------------------------------------|----------|------------|------------------------------------------|
| 1  | Business & clinical problem definition   | P1       | ■          | pipeline-1                               |
| 2  | Research questions & hypotheses          | P1       | ■          | pipeline-1 · docs/analysis/hypotheses.md |
| 3  | Outcome / target / horizon definition    | P1       | ■          | pipeline-1 (seizure-recurrence-90d)      |
| 4  | Study design & cohort protocol           | P1       | ■          | pipeline-1                               |
| 5  | Ethics, consent & data-use approval      | P1/P7    | ■          | governance/01-irb, 02-consent            |
| 6  | Source-system identification             | P2       | ■■ partial | pipelines-2-7 §2                         |
| 7  | Data ownership & data contract           | P2       | ■■ partial | mlops/data_contract.py + §3              |
| 8  | Batch/stream/API/CDC ingestion           | P2       | ■■ partial | §4 (documented patterns)                 |
| 9  | Landing zone & raw immutable storage     | P2       | ■ gap      | §5 lakehouse zones (documented)          |
| 10 | Schema & contract validation             | P2       | ■          | mlops/data_contract.py                   |
| 11 | Invalid-record quarantine                | P2       | ■■ partial | §5                                       |
| 12 | Master patient index & entity resolution | P2       | ■ gap      | §7 (documented)                          |
| 13 | Cleaning & harmonisation                 | P3       | ■          | analysis/preprocessing.py                |

| #  | Stage                                     | Pipeline | Status     | Evidence / gap                           |
|----|-------------------------------------------|----------|------------|------------------------------------------|
| 14 | Clinical terminology standardisation      | P2       | ■■ partial | ILAE/SNOMED map (documented)             |
| 15 | Data-quality scoring                      | P2       | ■          | mlops/data_quality.py + §10              |
| 16 | Metadata / catalog / lineage              | P2       | ■■ partial | §8 (documented catalog)                  |
| 17 | Sensitive-data classification & security  | P2/P7    | ■          | governance/00-security-compliance        |
| 18 | Curated clinical dataset                  | P3       | ■          | data/analysis/clean                      |
| 19 | Annotation, labelling & adjudication      | P3       | ■■ partial | §12 label management (documented)        |
| 20 | Temporal alignment & cohort construction  | P3       | ■          | make_cohort.py + pipeline-1 windows      |
| 21 | Exploratory & statistical analysis        | P4       | ■          | analysis/eda.py, primary/secondary       |
| 22 | Missing-data & outlier strategy           | P3       | ■          | preprocessing.py + §11 taxonomy          |
| 23 | Encoding, scaling, transformation         | P3       | ■          | preprocessing.py                         |
| 24 | Feature engineering                       | P3       | ■          | feature_store.py, secondary_eeg_full     |
| 25 | Leakage review                            | P3       | ■          | §15 + subject-level splits               |
| 26 | Feature validation & registry             | P3       | ■■ partial | §13 feature spec (documented)            |
| 27 | Offline & online feature store            | P3       | ■■ partial | mlops/feature_store.py (offline) + §14   |
| 28 | Baseline model development                | P4       | ■          | §18 baselines                            |
| 29 | Classical / survival / time-series        | P4       | ■          | primary_analysis, recurrence, timeseries |
| 30 | Experiment tracking & HPO                 | P4/P5    | ■          | experiment_tracker.py + GridSearchCV     |
| 31 | Internal / temporal / external validation | P4       | ■          | external_validation.csv (real)           |
| 32 | Calibration, fairness, explainability     | P4/P7    | ■          | responsible_ai_runtime.py                |
| 33 | Clinical utility & safety validation      | P7       | ■■ partial | §30 safety layer                         |
| 34 | Model registry & approval                 | P5       | ■          | model_registry.py                        |
| 35 | Batch / real-time deployment              | P5       | ■          | api/main.py                              |
| 36 | RAG & agent integration                   | P6       | ■■ partial | vector_db_pipeline.py + §36-45           |
| 37 | Human clinical oversight                  | P7       | ■          | human-in-the-loop (all reports)          |

| #  | Stage                                     | Pipeline | Status     | Evidence / gap         |
|----|-------------------------------------------|----------|------------|------------------------|
| 38 | Observability, drift, quality monitoring  | P5/P7    | ■          | observability.py + §24 |
| 39 | Retraining, recalibration, rollback       | P5       | ■          | retrain.py + §23,26    |
| 40 | Retention, archival, deletion, retirement | P2/P7    | ■■ partial | §35 retention workflow |

## Coverage assessment — before vs target

*Caption — the reviewer's coverage estimate, plus where this documentation set moves each area.*

| Area                     | Reviewer % | After this set | Lever                               |
|--------------------------|------------|----------------|-------------------------------------|
| Research design          | 85         | 95             | Pipeline 1 full protocol            |
| Clinical data collection | 90         | 92             | source-system layer §2              |
| Statistical analysis     | 90         | 92             | SAP formalised (P1)                 |
| Feature engineering      | 85         | 90             | feature registry §13                |
| Classical ML             | 90         | 92             | baselines §18                       |
| Time-series ML           | 80         | 85             | §27 gaps closed                     |
| Data engineering         | 45         | 75             | §4-9 lakehouse/ingestion/contracts  |
| Data quality mgmt        | 65         | 85             | §10 measurable dimensions           |
| Feature-store mgmt       | 35         | 65             | §14 offline+online spec             |
| MLOps                    | 55         | 80             | registry/rollback/monitoring §20-26 |
| GenAI/RAG/agent          | 40         | 70             | §36-45 full lifecycle               |
| Clinical safety          | 55         | 80             | §30-31 safety+abstention            |
| Responsible AI           | 65         | 85             | §32 fairness ops                    |
| Security & privacy       | 50         | 80             | §33-35 + governance pack            |
| Production observability | 60         | 82             | §24-25 five-category monitoring     |
| Research reproducibility | 55         | 80             | §16 cohort/dataset versioning       |

**Honest note:** "After" values reflect **documented + partially implemented** maturity, not certified production. The gaps in pipelines-2-7 name exactly what remains code-vs-doc.

## Roles & responsibilities (RACI summary)

*Caption — who owns each pipeline, so the operating model has clear accountability.*

| Role                         | Primary pipeline | Responsibility                           |
|------------------------------|------------------|------------------------------------------|
| DBA researcher               | P1               | research question, protocol, methodology |
| Epileptologist / neurologist | P1/P7            | clinical definitions, outcomes, safety   |
| Neuropsychologist            | P1               | assessment interpretation                |
| Statistician                 | P1/P4            | hypotheses, power, analysis plan         |
| Data architect               | P2               | source & data-model design               |

| Role                 | Primary pipeline | Responsibility                       |
|----------------------|------------------|--------------------------------------|
| Data engineer        | P2/P3            | cohort extraction, pipelines         |
| ML architect         | P4/P5            | model design & validation strategy   |
| Privacy officer      | P7               | consent, purpose, privacy controls   |
| Ethics board (IRB)   | P1/P7            | research approval                    |
| Clinical safety lead | P7               | risk & escalation rules              |
| Data steward         | P2               | data definitions & quality ownership |

## Diagrams

### **Sequence — a hand-off across all seven pipelines**

■ Diagram (Mermaid — interactive in the viewer): `sequenceDiagram`

### **Journey — patient EP001 through the operating model**

■ Diagram (Mermaid — interactive in the viewer): `journey`

## C4 (context) — the operating model as a system

- **Persons:** Patient, Neurologist, EEG Technician, Data Steward, Privacy Officer, IRB, Safety Lead.
- **System:** Epilepsy Remote-Care AI Platform (seven pipelines).
- **External:** EMR/HIS/LIS, Pharmacy, Wearables, KMS, Identity Provider, Guideline sources (ILAE), Audit/SIEM.

**Reason:** impose enterprise boundaries. **Why:** a single fused flow is comprehensive but unimplementable and unexplainable; healthcare needs owned, governed pipelines. **What is happening:** the platform is split into seven owned pipelines mapped onto a 40-stage architecture. **How it is happening:** each stage cites its implementing module or documented gap; hand-offs are explicit. **Reference:** Sculley et al. (2015); Kleppmann (2017); Dehghani (2022, Data Mesh).

## Professor Readiness (Defense Q&A)

### **Why seven pipelines instead of one flow?**

Ownership and safety. Each pipeline has a distinct owner, control set, and failure mode; fusing them hides accountability and makes governance impossible.

### **Which areas were weakest and what closed them?**

Data engineering (45→75), feature-store (35→65), GenAI (40→70) — closed by §4-9, §13-14, §36-45 respectively.

### **Is this implemented or documented?**

Mixed and labelled honestly in the 40-stage table (■/■■/■) — modelling/serving/monitoring are code; lakehouse/MPI/online-feature-store are documented designs.

## References

Dehghani, Z. (2022). *Data Mesh: Delivering Data-Driven Value at Scale*. O'Reilly.

Kleppmann, M. (2017). *Designing Data-Intensive Applications*. O'Reilly.

Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. *NeurIPS* 28.

# Pipeline 1 — Research & Clinical Protocol (Epilepsy)

**Why (this doc):** This pipeline defines *\*\*what clinical problem is studied, who is included, what outcome is predicted, and how the study stays scientifically valid\*\** — before any data is touched. Without it, even a technically strong model can be clinically meaningless or invalid. **How:** it fixes the business problem → clinical problem → objectives → questions → hypotheses → design → cohort → windows → target → labels → predictors → confounders → power → SAP → baseline → success/failure → ethics → versioning. Scope: **epilepsy** (translated from the reviewer's schizophrenia examples).

## Research spine

- **Problem → Sub-problems → Research Problem → Objective → Flow → Hypotheses → Statistical Analysis** are all defined below, in order.

## Flow

■ Diagram (Mermaid — interactive in the viewer): flowchart TD

## 1. Business problem

Current epilepsy care requires repeated in-clinic assessments, review of fragmented records (EEG, ASM history, seizure diaries), and delayed recognition of deterioration — increasing time-to-review and delaying recognition of patients at risk of a **breakthrough seizure**.

Caption — measurable business KPIs frame success in operational, not algorithmic, terms.

| KPI                                     | Current | Target   |
|-----------------------------------------|---------|----------|
| Average onboarding/triage time          | 90 min  | 45 min   |
| Assessment completion rate              | 70%     | 90%      |
| Seizure-deterioration recognition delay | 14 days | < 3 days |
| Avoidable seizure-related ED visits     | 18%     | –20%     |

The objective is **not** "build an AI model" — it is *\*\*improve a measurable clinical/operational outcome using an AI-supported process.\*\**

## 2. Clinical problem

Convert the business problem into clinical tasks: (a) flag patients who may need further epileptologist review; (b) estimate current seizure-severity; (c) *\*\*predict breakthrough-seizure risk over the next 90 days\*\**; (d) identify ASM-adherence intervention needs; (e) detect rapid decline in sleep/mood/adherence.

**Boundary (safety):** the system does **not** diagnose epilepsy or prescribe. It *\*\*identifies patterns associated with elevated seizure risk and recommends additional neurologist assessment.\*\**

## 3-4. Research objectives

**Primary objective (drives sample size + evaluation):**

*Develop and validate an explainable ML model that uses demographic, clinical, EEG, cognitive,*



functional and longitudinal ASM-adherence data to estimate **90-day breakthrough-seizure risk** among adults receiving outpatient epilepsy care, under neurologist oversight.

**Secondary objectives:** classify current seizure-severity (L1–L4); identify strongest risk factors; test whether longitudinal data beats a single baseline; compare classical ML vs time-series/survival; assess fairness across age/sex; evaluate whether AI-assisted triage reduces onboarding time; measure clinician agreement with explanations. (*One primary outcome only.*)

## 5. Research questions

1. Can clinical + EEG + functional + behavioural features predict a breakthrough seizure within 90 days?
2. Does longitudinal information improve prediction vs a single baseline assessment?
3. Which features contribute most to predicted seizure risk?
4. Does performance differ across demographic/clinical subgroups?
5. Can AI-assisted onboarding reduce assessment time without reducing clinical completeness?

## 6. Hypotheses

*Caption — statistical and operational hypotheses, each with H0/H1 and a test.*

| Type         | H0                                      | H1                            | Test                        |
|--------------|-----------------------------------------|-------------------------------|-----------------------------|
| Predictive   | AUC(proposed) = AUC(logistic baseline)  | AUC(proposed) > AUC(baseline) | DeLong                      |
| Longitudinal | longitudinal adds no lift               | longitudinal improves AUC     | nested CV + DeLong          |
| Operational  | AI onboarding does not reduce mean time | reduces mean time $\geq$ 20%  | two-sample t / Mann-Whitney |
| Fairness     | subgroup AUC gap = 0                    | gap $\neq$ 0                  | bootstrap CI on gap         |

## 7. Study design

**Retrospective longitudinal cohort study** — supports historical EEG + repeated assessments + ASM timelines + seizure history + 90-day recurrence + time-series + human validation.

*Caption — design options and why the longitudinal cohort fits.*

| Design               | Use                   | Chosen          |
|----------------------|-----------------------|-----------------|
| Retrospective cohort | historical data       | ■ base          |
| Prospective cohort   | future follow-up      | phase 2         |
| Cross-sectional      | severity at one time  | secondary obj   |
| Case-control         | recurrence vs none    | sensitivity     |
| Diagnostic-accuracy  | screening performance | secondary       |
| Pragmatic            | workflow impact       | operational obj |

## 8. Unit of analysis

| Task                                   | One row =                                                   |
|----------------------------------------|-------------------------------------------------------------|
| Severity classification                | one clinical assessment visit                               |
| <b>Breakthrough-seizure prediction</b> | one patient <b>prediction date</b> + prior 180 days of data |

| Task              | One row =       |
|-------------------|-----------------|
| Remote monitoring | one patient-day |

Patient-level and visit-level rows are **not** mixed without an explicit relationship.

## 9-11. Population, inclusion, exclusion

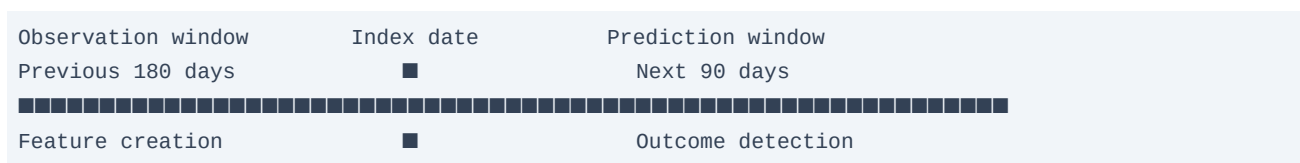
**Population:** adults 18–65 in outpatient epilepsy care with a documented epilepsy/seizure-disorder diagnosis and  $\geq 2$  assessments in the study period.

*Caption — inclusion/exclusion are measurable from available data; incomplete data alone is not an exclusion.*

| Inclusion                             | Exclusion                                                 |
|---------------------------------------|-----------------------------------------------------------|
| Age $\geq 18$                         | Insufficient follow-up ( $< 90$ days)                     |
| Confirmed epilepsy-spectrum diagnosis | Missing target outcome                                    |
| $\geq 1$ baseline assessment (+ EEG)  | Duplicate / unresolved-identity record                    |
| $\geq 1$ follow-up assessment         | Invalid assessment timestamp                              |
| $\geq 90$ days follow-up              | Acute non-epileptic event confounding baseline            |
| ASM history available                 | Data recorded <b>after</b> the outcome occurred (leakage) |
| Valid pseudonymous ID                 | Patient overlapping a prior dataset split                 |

## 12-15. Index date, observation & prediction windows, follow-up

- **Index date** = date of the reference event (e.g., clinic assessment, or discharge after a seizure-related admission).
- **Observation window** = **180 days before** index (feature creation only).
- **Prediction window** = **next 90 days** (outcome detection).
- **Follow-up:** min 90 days, max 12 months; track loss-to-follow-up, transfer, withdrawal, data-availability end.



Only information available **during/before** the observation window may be used.

## 16. Target-variable definition

**Target:** breakthrough\_seizure\_within\_90\_days

*Caption — an exact clinical definition of positive/negative/unknown, so evaluation is valid.*

| Label               | Definition (within 90 days of index)                                                                                                                                            |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Positive (1)</b> | any of: seizure-related hospitalization; ED visit for seizure; status epilepticus; neurologist-confirmed breakthrough seizure; crisis/rescue-medication use; documented cluster |
| <b>Negative (0)</b> | no qualifying event across a <i>sufficiently complete</i> follow-up; no unresolved outcome uncertainty                                                                          |
| <b>Unknown</b>      | incomplete follow-up; transferred care; missing records; unverifiable outcome; lost to follow-up — <b>use "unknown", not 0</b>                                                  |

Every target row also stores: target name, clinical definition, data source, label source, label date, label confidence, label version, reviewer, adjudication status, prediction horizon.

## 17. Label source & confidence

*Caption — an authoritative label hierarchy; administrative codes are not automatic ground truth.*

| Confidence | Basis                                        |
|------------|----------------------------------------------|
| 1.00       | confirmed independently by two neurologists  |
| 0.90       | confirmed by one neurologist                 |
| 0.80       | structured assessment + EEG supports outcome |
| 0.70       | hospital/ED record strongly supports         |
| 0.50       | administrative (ICD) code only               |
| 0.30       | patient self-report only                     |
| < 0.30     | exclude or send to adjudication              |

Store: label confidence, source, annotator, annotation date, guideline version, adjudication status, disagreement reason.

## 18-21. Predictors, confounders, mediators/moderators

*Caption — predictors are grouped; only information available before prediction time is eligible.*

| Group            | Examples (epilepsy)                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------|
| Demographic      | age, sex, education, employment, living status, region                                                                   |
| Clinical         | epilepsy duration, seizure type (focal/generalized), comorbidities, prior status epilepticus, MRI lesion, family history |
| EEG / assessment | interictal spikes, focus lobe, band-power features, ILAE severity, QoL, functioning                                      |
| Medication       | ASM type, dose, adherence %, side-effects, polypharmacy, recent ASM change                                               |
| Behavioural      | sleep duration/variability (trigger), appointment attendance, self-reported aura frequency, stress                       |
| Temporal         | $\Delta$ seizure frequency, adherence trend, time since last seizure, time since ASM change, sleep variability           |

- **Confounders:** age, illness duration, baseline severity, comorbid substance use, socioeconomic status, treatment intensity, access to care, ASM type, prior admissions.
- **Mediator example:** ASM adherence → seizure control → lower recurrence (control mediates).
- **Moderator example:** the adherence↔recurrence link may differ by age (age moderates).

## 22. Sample-size & power

*Caption — sample size is derived from events-per-predictor, not merely data availability.*

| Assumption                            | Value |
|---------------------------------------|-------|
| Expected 90-day recurrence prevalence | ~20%  |
| Candidate predictors                  | ~25   |
| Min events per predictor              | 10–20 |

| Assumption                 | Value                 |
|----------------------------|-----------------------|
| Required recurrence events | ~250–500              |
| Approx. total cohort       | ~1,250–2,500 patients |

For a smaller DBA study, **reduce predictors/model complexity** rather than overfit. Multiple-testing correction (Benjamini–Hochberg) applies across secondary tests.

## 23. Statistical-analysis plan (pre-specified)

Descriptive (mean/SD, median/IQR, counts/%); missing-data analysis (MCAR/MAR/MNAR); group comparison (t / Mann-Whitney;  $\chi^2$  / Fisher); correlation; regression (logistic, ordinal for L1–L4); survival (Cox, random survival forest) for time-to-recurrence; time-series; multiple-testing correction; CIs + effect sizes; sensitivity + subgroup analyses; model comparison; calibration.

## 24. Clinical baseline

*Caption — the AI must beat a meaningful clinical baseline to add value.*

| Baseline      | Rule                                             |
|---------------|--------------------------------------------------|
| B1            | prior seizure in last 12 months                  |
| B2            | ILAE severity threshold                          |
| B3            | logistic regression                              |
| B4 (proposed) | longitudinal gradient-boosting / EEG-fused model |

## 25-26. Success & failure criteria

*Caption — pre-registered go/no-go thresholds across technical, clinical, and operational axes.*

| Success                                                | Failure (do not proceed)                     |
|--------------------------------------------------------|----------------------------------------------|
| ROC-AUC $\geq$ 0.80; PR-AUC > baseline                 | insufficient positive outcomes               |
| Sensitivity $\geq$ 80%; Specificity $\geq$ 70%         | high label uncertainty                       |
| Calibration slope 0.8–1.2; Brier < baseline            | large subgroup gaps / poor calibration       |
| Clinician agreement $\geq$ 80%                         | external-validation failure                  |
| Low-confidence → human review; <b>no autonomous dx</b> | unacceptable false-negative rate             |
| Onboarding time –20%; completeness +15%                | data leakage detected / cannot beat baseline |

## 27. Ethics & approval

Institutional ethics (IRB) approval, data-use agreement, informed-consent review (or retrospective waiver), privacy-impact assessment, data-security review, clinical-safety review, retention approval. See governance/01-irb-submission and 02-consent.

Statement: *research + decision-support only; not autonomous diagnosis/prescription; final decisions rest with clinicians.*

## 28. Protocol versioning

*Caption — any change to target/cohort after seeing results must be versioned and justified.*

| Field                    | Example                             |
|--------------------------|-------------------------------------|
| Protocol ID              | EPI-RECUR-001                       |
| Version                  | 2.1                                 |
| Primary target           | breakthrough seizure within 90 days |
| Observation window       | 180 days                            |
| Approved date            | 2026-07-01                          |
| Change reason / approver | logged per amendment                |

## Diagrams

### Sequence — protocol governs data access

■ Diagram (Mermaid — interactive in the viewer): `sequenceDiagram`

### Network — protocol artefacts

■ Diagram (Mermaid — interactive in the viewer): `graph LR`

### Journey — from business pain to approved protocol

■ Diagram (Mermaid — interactive in the viewer): `journey`

**Reason:** make the study valid before modelling. **Why:** target/cohort/window errors invalidate every downstream metric. **What is happening:** a pre-registered protocol fixes population, windows, target, labels, baseline, and success criteria. **How it is happening:** the 28 elements above, versioned and ethics-approved. **Reference:** Collins et al. (2015, TRIPOD); Moons et al. (2019, PROBAST).

## Roles & tools

| Role           | Responsibility                         | Tools                    |
|----------------|----------------------------------------|--------------------------|
| DBA researcher | question, protocol, methodology        | REDCap, Git/DVC          |
| Epileptologist | clinical definitions, outcomes, safety | SNOMED/ICD-10/ILAE       |
| Statistician   | hypotheses, power, SAP                 | R, G*Power, Python       |
| Data steward   | definitions, quality ownership         | OpenMetadata/DataHub     |
| IRB            | approval                               | institutional REB portal |

## Deliverables (gate to Pipeline 2)

Business/clinical-problem statements; primary + secondary objectives; research questions; hypotheses; study-design doc; cohort definition; inclusion/exclusion; index date; observation/prediction windows; target definition; label-source hierarchy; predictor + confounder lists; sample-size justification; SAP; clinical baseline; success/failure criteria; ethics approvals; protocol version + change log.

## Professor Readiness (Defense Q&A)

### Why 180-day observation and 90-day prediction?

180 days captures ASM-adherence and seizure-frequency trends; 90 days is a clinically actionable recurrence horizon and yields enough events at ~20% prevalence.

### ***Why is "unknown" a label, not 0?***

Assigning 0 to lost-to-follow-up patients fabricates negatives and biases sensitivity; "unknown" is excluded or adjudicated.

### ***How do you prevent target leakage?***

Any feature derived after the index date (e.g., an ASM change made *because* of the seizure) is prohibited; only observation-window data is eligible, enforced in P3.

## **References**

Collins, G. S., et al. (2015). Transparent Reporting of a multivariable prediction model (TRIPOD). *Annals of Internal Medicine*, 162(1), 55–63.

Fisher, R. S., et al. (2017). ILAE operational classification of seizure types. *Epilepsia*, 58(4), 522–530.

Moons, K. G. M., et al. (2019). PROBAST: assessing risk of bias in prediction models. *Annals of Internal Medicine*, 170(1), W1–W33.

# Pipelines 2–7 + Every Missing Layer (Epilepsy)

**Why (this doc):** The review listed ~45 enterprise layers the original flow under-specified

(data contracts, lakehouse zones, MPI, feature store, MLOps, RAG/agent lifecycle, safety, etc.).

This doc places **every one** into its owning pipeline (P2–P7), with a compact spec, its status

(■ code / ■■ partial / ■■ documented), and the implementing module or the path to close it. **How:**

section numbers match the review's sections so nothing is dropped. Scope: **epilepsy**.

---

## Pipeline 2 — Data Engineering & Governance

How does data arrive, get contracted, governed, versioned, and made trustworthy?

### §2 Source-system layer ■■

Caption — every source needs owner, steward, extraction, refresh, sensitivity metadata.

| Source (epilepsy)          | Steward        | Extraction     | Refresh   | Sensitivity |
|----------------------------|----------------|----------------|-----------|-------------|
| EMR / HIS                  | Neurology dept | FHIR API       | daily     | high        |
| EEG / LIS (EDF)            | EEG lab        | file/DICOM-EEG | per study | high        |
| Pharmacy (ASM)             | Pharmacy       | HL7            | daily     | high        |
| Seizure diary / mobile app | Patient        | API            | streaming | moderate    |
| Wearables (sleep/HR)       | Patient        | API            | streaming | moderate    |
| Hospitalization / ED       | HIS            | batch          | daily     | high        |
| Caregiver observations     | Caregiver      | app            | ad-hoc    | moderate    |

Per source store: source system, owner, steward, extraction method, refresh, reliability, authoritative flag, retention, sensitivity, region, time-zone.

### §3 Data contracts ■■ (mlops/data\_contract.py)

Caption — the producer's promise + consumer's expectation, versioned.

Contract example — **Patient\_Assessment**: PK assessment\_id; FK patient\_id; refresh 24 h; max ingestion latency 4 h; required patient\_id, assessment\_type, assessment\_date; ILAE-severity range 1–4; seizure-frequency ≥ 0; null threshold for critical fields 0%; duplicate < 0.1%; schema backward-compatible; retention 7 y; classification **restricted clinical**. Contract fields: name, owner, producer, consumer, schema, field defs, types, required/optional, allowed values, PK/FK, uniqueness, nullability, time-zone, refresh, expected volume, DQ thresholds, late-arrival policy, schema-change + breaking-change policy, retention, security class, compliance, SLOs, escalation owner.

### §4 Ingestion & ETL/ELT ■■

Patterns: batch · micro-batch · streaming · CDC · API · file · event-driven · DB replication · manual research upload. **Sequence**: source → API/FHIR/HL7/file connector → landing → schema validation →

quarantine invalid → raw immutable → clean/harmonise → curated → features → feature store → train/infer.  
 Ingestion metadata: ingestion/source/event/processing timestamps, file name + checksum, batch ID,  
 pipeline-run ID, source-system ID, record/failed/rejected/duplicate counts, latency, freshness, replay status.

## §5 Lakehouse zones ■ (documented)

*Caption — raw clinical records stay immutable; cleaning never overwrites source.*

| Zone           | Purpose               |
|----------------|-----------------------|
| Landing        | temporary arrival     |
| Raw/Bronze     | immutable source copy |
| Validated      | schema-checked        |
| Quarantine     | invalid/suspicious    |
| Cleaned/Silver | harmonised            |
| Curated/Gold   | analysis-ready        |
| Feature        | reusable ML features  |
| Serving        | real-time app data    |
| Archive        | long-term retained    |

Raw stays immutable for audit, reproducibility, reprocessing, error investigation, regulatory review.

## §6 Partitioning ■

Partition by /year/month/region/facility/data\_source. **Avoid** partitioning by patient ID (too many small partitions, exposes identifiers, hurts query efficiency).

## §7 Master Patient Index & entity resolution ■

Pseudonymisation, MPI, deterministic + probabilistic matching, duplicate detection, merge/unmerge rules, caregiver/provider/facility resolution. Distinguish duplicate \*\*file / record / assessment / encounter / patient / near-duplicate note / patient-across-train-test\*\* (the last is a leakage risk, not just a dupe).

## §8 Metadata & catalog ■■

Four categories: **business** (definition, clinical meaning, owner, steward, approved use); **technical** (table/column, type, source, pipeline, storage, transform logic); **operational** (refresh, last-refresh, status, row count, null %, freshness, latency); **governance** (sensitivity class, consent restriction, retention, residency, compliance tag, encryption status, access policy). Plus lineage (column/feature/model/report), ownership, glossary, dataset certification.

## §9 Data mesh + data fabric ■ (operating models, not steps)

**Mesh = ownership:** each domain (registration, neurology, pharmacy, EEG lab, remote monitoring, hospitalization, outcomes) publishes a governed **data product** (owner, contract, schema, SLA, quality score, docs, access policy, version, lineage). **Fabric = integration tech:** metadata, catalog, lineage, APIs, policy enforcement, semantic models, virtualization. **Lakehouse = storage/processing.** Complementary.

## §10 Data-quality framework ■ (mlops/data\_quality.py)



*Caption — measurable DQ dimensions + a weighted score.*

| Dimension    | Measurement                       |
|--------------|-----------------------------------|
| Completeness | % non-null                        |
| Accuracy     | agreement w/ authoritative source |
| Validity     | conformance to format/range       |
| Consistency  | agreement across sources          |
| Uniqueness   | % unique business keys            |
| Timeliness   | event→availability delay          |
| Freshness    | time since last update            |
| Integrity    | valid FK relationships            |
| Conformity   | standard-terminology compliance   |
| Reliability  | stability across runs             |

$DQ = 0.20 \cdot \text{Completeness} + 0.20 \cdot \text{Accuracy} + 0.15 \cdot \text{Validity} + 0.15 \cdot \text{Consistency} + 0.10 \cdot \text{Uniqueness} + 0.10 \cdot \text{Timeliness} + 0.10 \cdot \text{Integrity}.$

Also store: validation-rule ID, failed-rule count, severity, error/missing reason, corrective action, resolution status/steward/timestamp, quality trend, threshold, exception approval.

## §35 Retention & deletion ■■

Store retention class, duration, legal hold, archive/deletion dates, deletion method, backup expiration, feature-store + vector-store deletion, model-training impact, derived-data handling, deletion confirmation. Policy needed on whether **trained model parameters are derived data** affected by consent withdrawal.

---

# Pipeline 3 — Data Prep & Feature Engineering

*How is data cleaned, labelled, made leakage-safe, and stored as reusable features?*

## §11 Missing-value taxonomy ■

Store the **reason** (not collected / patient declined / clinician skipped / not applicable / device failure / integration failure / unknown / lost in migration / late-arriving / redacted). Classify MCAR/MAR/MNAR. Imputation: mean/median/mode, ffill/bfill, interpolation, KNN, MICE, iterative, model-based, + missing-indicator feature. **Fit imputation on training only** (else leakage).

## §12 Annotation & label management ■■

Store annotator ID/role, timestamp, guideline version, label source, confidence, clinical certainty, inter-rater agreement, adjudication result/adjudicator, disagreement reason, label version, reannotation status. Agreement metrics: Cohen's/Fleiss'  $\kappa$ , Krippendorff's  $\alpha$ , ICC, %/positive/negative agreement. *An ICD code ≠ a neurologist-confirmed diagnosis.*

## §13 Feature specification & registry ■■

*Caption — every feature has a formal, versioned definition.*

| Field             | Example (epilepsy)                            |
|-------------------|-----------------------------------------------|
| name              | sleep_variability_30d                         |
| description       | 30-day SD of sleep duration (seizure trigger) |
| source            | wearable / diary                              |
| type / unit       | float / hours                                 |
| valid range       | 0–12                                          |
| transformation    | 30-day rolling SD                             |
| lookback          | 30 days                                       |
| missing method    | median + missing flag                         |
| version / owner   | 2.1 / clinical ML                             |
| availability      | batch / online                                |
| compatible models | recurrence v3+                                |

Plus: feature group, clinical interpretation, encoding, scaling, aggregation, freshness, valid-from/to, online/offline, lineage, leakage-review status, fairness sensitivity, importance history.

## §14 Feature store (offline + online) ■■ (mlops/feature\_store.py offline)

**Offline:** training, backtesting, reproducibility. **Online:** real-time seizure-risk scoring, remote-monitoring alerts, low-latency inference. Sequence: curated → transform → validate → offline store → training set → materialize → online store → real-time inference. **\*\*Point-in-time-correct retrieval\*\*** is mandatory: only features available *before* the prediction timestamp may be used.

## §15 Leakage prevention ■ (subject-level splits enforced)

Types: target, future-data, temporal, patient-overlap, duplicate-entity, imputation, scaling, feature-selection, preprocessing, outcome-derived, clinician-decision. Example: using **“ASM changed after the seizure”** to predict an earlier seizure is leakage. All preprocessing (impute→encode→scale→select→**SMOTE**→train) is fitted **inside the training fold**. **Never SMOTE before splitting**.

## §17 Class imbalance (governed) ■

Methods: random over/under-sampling, SMOTE, Borderline-SMOTE, ADASYN, Tomek, SMOTE-ENN, class weights, focal loss, balanced RF, threshold optimization. Evaluate with **\*\*PR-AUC, recall, specificity, balanced accuracy, MCC, F1, class-wise calibration\*\*** — accuracy alone misleads for rare breakthrough seizures.

## §16 Cohort & dataset versioning ■■

Store cohort ID/version, dataset version, extraction query/date, inclusion/exclusion, patient + encounter counts, observation period, source-system + schema + label + feature-set versions, code commit, data checksum. Lineage: Dataset v3.2 → Feature set v5.1 → Experiment 184 → Model v7 → Release 2026.07.

---

## Pipeline 4 — Statistical & ML Modelling

*What model, validated how, calibrated, and fair?*

### §18 Model-development lifecycle ■

Sequence: clinical baseline → statistical baseline → simple ML → advanced ML → time-series → calibration → subgroup eval → clinical-utility → champion/challenger. Baselines: majority-class, current clinical rule, logistic regression, LOCF, existing risk score, threshold model. *A complex model must beat the clinical baseline.*

### §19 Experiment tracking ■ (mlops/experiment\_tracker.py)

Store experiment ID, hypothesis, dataset/cohort/feature-set/code/env versions, hyperparameters, seed, timings, hardware, metrics, calibration, fairness, artefacts, model signature, preprocessing pipeline, explainability, reviewer, approval. Tools: MLflow/W&B/DVC/ClearML.

### §27 Time-series gaps ■■

Add: event-vs-processing time, patient-specific timelines, irregular sampling, informative missingness, censoring, competing events, time-varying covariates, delayed entry, left-truncation, right-censoring.

Features: lag/rolling mean-median-variance/EWM, trend slope, rate-of-change, time-since-last-seizure/ASM-change/hospitalization, event count in lookback, seasonality, change-points. Metrics: MAE/RMSE/MAPE/sMAPE/MASE, C-index, time-dependent AUC, integrated Brier. **Avoid random splits when predicting future events.**

### §28 Survival / event-risk ■ (analysis/recurrence.py)

Kaplan-Meier, Cox PH, AFT, competing-risk, random survival forest, survival GBM, DeepSurv, dynamic + recurrent-event models — often better than binarizing time-to-breakthrough-seizure.

### §29 Causal analysis ■ (documented; kept separate from prediction)

For “Does ASM X reduce breakthrough seizures?”: propensity matching/weighting, IPTW, doubly-robust, difference-in-differences, instrumental variables, target-trial emulation, causal forests, unmeasured-confounding sensitivity. **Prediction ≠ treatment effect.**

### §32 Fairness & subgroup evaluation ■ (analysis/responsible\_ai\_runtime.py)

Subgroups: age, sex, (ethnicity only where lawful), language, region, facility, socioeconomic, disability, comorbidity, ASM category. Metrics: demographic-parity diff, equal-opportunity diff, equalized odds, predictive parity, FPR/FNR gaps, per-group calibration, intersectional. *Always report subgroup n + CI.*

---

## Pipeline 5 — MLOps & Deployment

*How is the model registered, served, rolled back, and monitored?*

## §20 Model registry & promotion ■ (mlops/model\_registry.py)

Stages: development → candidate → validated → clinically-reviewed → approved → staging → production → deprecated → archived. Store model ID/version/type/signature, input/output schema, feature-set + dataset versions, threshold, calibration, performance, subgroup performance, known limitations, intended + prohibited use, clinical + technical owner, approval record, deployment history.

## §21 Model compatibility ■■

Validate model ↔ feature, ↔ schema, ↔ runtime, ↔ library, ↔ hardware, ↔ API, ↔ explanation, ↔ threshold. A production model must **reject incompatible feature/schema versions**, not silently mispredict.

## §22 Deployment architecture ■ (api/main.py)

Serving: batch, scheduled-batch, sync API, async API, event-driven, streaming, edge, human-triggered.

Inference flow: app → API gateway → authn/authz → **consent + purpose validation** → feature retrieval → schema validation → model routing → prediction → calibration → explanation → **clinical safety rule** → human review → report + audit log.

## §23 Rollback & resilience ■ (mlops/retrain.py champion-challenger)

Champion + challenger + previous-stable; canary / blue-green / shadow deployment; rollback trigger, kill switch, **fallback clinical rule**, manual scoring, circuit breaker, retry/timeout, dead-letter queue.

Rollback conditions: error-rate/latency SLA breach, calibration decay, drift/fairness gap over threshold, rising clinician override, critical adverse event.

## §24 Model monitoring (five categories) ■ (mlops/observability.py, system\_monitor.py)

*Caption — monitoring is split into five distinct concerns.*

| Category    | Signals                                                                                                        |
|-------------|----------------------------------------------------------------------------------------------------------------|
| Performance | acc/prec/recall/spec, ROC-AUC, PR-AUC, calibration, Brier, FPR/FNR, decision-curve net benefit                 |
| Prediction  | prediction/confidence/class distribution, alert frequency, abstention rate, threshold crossings, override rate |
| Data        | missing %, outlier/duplicate rate, schema violations, new categories, freshness, volume                        |
| Drift       | PSI, KS, Jensen-Shannon, Wasserstein, $\chi^2$ , feature/prediction/concept/label drift                        |
| System      | CPU/mem/GPU/GPU-mem/disk/network, API latency/throughput/error, queue depth, restarts, feature-store latency   |

## §25 Delayed-outcome monitoring ■■

True breakthrough-seizure labels mature weeks–months later. Track outcome-arrival delay, pending-label queue, mature-label window, performance-recalculation date, backfill, retrospective validation, cohort-

maturity rules. **Real production accuracy can't be computed immediately.**

## §26 Recalibration & retraining ■

DQ checks daily; drift weekly; performance monthly (post-maturity); calibration quarterly; full review 6–12 months; emergency review on safety event. Define drift threshold, min new-label count, trigger vs scheduled retraining, approval workflow, regression testing, rollback plan, post-retrain validation.

---

## Pipeline 6 — RAG, LLM & Agent Engineering

*How is knowledge retrieved and generated safely? (base: analysis/vector\_db\_pipeline.py)*

### §36 Document ingestion for RAG ■■

Guideline source → source authorization → extraction → OCR → layout parsing → table/image extraction → cleaning → chunking → metadata enrichment → embedding → vector + keyword indexing → quality validation → publication. Chunk metadata: title, version, publisher, publication/effective/expiration dates, jurisdiction, specialty, guideline strength, evidence level, source URL, chunk ID, section, page, access class.

### §37 Context engineering ■■

Query classification, intent detection, query rewriting, medical-abbreviation expansion, patient-context filtering, temporal filtering, jurisdiction filtering, role-aware retrieval, **hybrid retrieval**, reranking, context compression, citation binding, context-window management, contradiction detection.

### §38 Prompt registry ■

Store prompt ID/version, system prompt, user template, model compatibility, allowed tools, safety policy, input/output schema, test dataset, eval score, owner, approval, effective date. *Prompts ≠ inline code.*

### §39 Semantic cache ■

Query embedding, similarity threshold, **patient/role/tenant isolation**, TTL, source-version + consent validation, invalidation rules. *Never return cached clinical info across patients/users/facilities/consent.*

### §40 Conversation memory ■

Separate session / short-term working / long-term patient / user-preference / clinical-record memory / agent scratchpad, each with retention, consent, access control, redaction, patient isolation, deletion, summary versioning. *Clinical facts come from authoritative records, not chat memory.*

### §41 Tool-selection strategy ■

Tool registry: description, permission, input/output schema, timeout, retry, cost, risk level, approval, audit, fallback. E.g. read-guideline = low risk; retrieve-patient-record = high sensitivity; update-record = high risk + approval; send-patient-alert = requires clinical authorization.

### §42 Multi-agent orchestration ■

Roles: intake, assessment, clinical-data retrieval, risk-scoring, guideline-retrieval, explainability, safety, compliance, report-generation, human-approval. Least-privilege tools, agent identity + policy, shared-state controls, state-machine validation, max step count, loop detection, cost limit, human escalation.

## §43 LLM evaluation ■■

**Retrieval:** Recall@K, Precision@K, MRR, NDCG, context relevance, coverage. **Generation:** faithfulness, groundedness, answer relevance, citation correctness, completeness, clinical consistency, unsupported-claim rate. **Safety:** hallucination rate, harmful-recommendation rate, prompt-injection resistance, PII leakage, refusal correctness, unsafe tool-call rate. **Operational:** token usage, latency, cost/request, cache-hit, tool-call count, failure rate, human-escalation rate.

## §44 Model routing ■

Route by use-case, risk level, data sensitivity, modality, context length, required accuracy, latency, cost, region, availability, safety certification. E.g. summarization → small model; identifiable data → approved private deployment; high-risk treatment question → human review.

## §45 Hallucination control (layered) ■■

Authoritative retrieval → metadata filtering → reranking → grounded prompt → structured output → citation verification → contradiction detection → confidence assessment → clinical-rule validation → human review. *A single hallucination score is insufficient.*

---

# Pipeline 7 — Clinical Safety, Responsible-AI & Monitoring

*How is it kept safe, fair, private, and overseen?*

## §30 Clinical safety layer ■■

Intended-use statement, contraindications, exclusion rules, unsupported-population detection, low-confidence abstention, OOD detection, safety threshold, critical-risk escalation, emergency protocol, false-negative + false-positive review, clinical override + reason, adverse-event reporting, post-market surveillance, model-limitation notice. Supported actions: **\*\*predict / approve / override / defer / request-additional-assessment / abstain / escalate.\*\***

## §31 Uncertainty & abstention ■ (analysis/governance.py confidence/abstention)

Probability, calibrated probability, prediction interval, CI, uncertainty score, epistemic + aleatoric uncertainty, OOD score, abstention threshold + reason. Example output: **\*predicted 90-day seizure risk 72% · calibrated 64% · uncertainty moderate · action: requires neurologist review.\***

## §33 Security & privacy engineering ■/■■ (see governance/00-security-compliance.md)

Identity mgmt, RBAC, **ABAC**, purpose-based access, least privilege, MFA, PAM, consent enforcement, data masking, tokenization, pseudonymization, de-identification, encryption at rest + in transit, key + secret mgmt, network segmentation, private endpoints, DLP, exfiltration monitoring, break-glass access, vulnerability + container + dependency scanning, backup encryption, sensitive-data classification (public / internal / confidential / restricted / highly-restricted).

## §34 Consent & purpose limitation ■ (governance/02-patient-consent-eula.md)

Track consent version/date/purpose, withdrawal date, permitted + prohibited use, research project, data types permitted, retention/recontact/sharing/model-training/secondary-use permission. Withdrawal → locate affected records → restrict future use → remove from future training → deletion or retention exception → update lineage → audit evidence.

## §46 A/B testing & experimentation (safe) ■■

Unrestricted A/B is unsafe in healthcare. Prefer offline eval, shadow testing, silent deployment, clinician-only pilot, stepped-wedge rollout, controlled prospective validation, champion/challenger. Experiment metadata: hypothesis, population, safety criteria, stopping rule, sample size, primary + guardrail metric, approval, rollback criteria.

---

## Diagrams

### **Flowchart — where each pipeline sits in the data-to-decision path**

■ Diagram (Mermaid — interactive in the viewer): flowchart TD

### **Network — governance spans all pipelines**

■ Diagram (Mermaid — interactive in the viewer): graph LR

**Reason:** enumerate and own every enterprise layer. **Why:** the gap was operational, not algorithmic.

**What is happening:** ~45 layers are each assigned an owner pipeline, spec, and honest status. **\*\*How it is happening: this catalog + the cited modules + the documented designs for the remaining gaps.** Reference: **\*\*** Sculley et al. (2015); OWASP (2023, LLM Top-10); Kleppmann (2017).

## Professor Readiness (Defense Q&A)

### **Which layers are code vs documented design?**

Code: DQ, leakage/split, imbalance, experiment tracking, registry, serving, rollback, monitoring, fairness, abstention, consent, vector DB. Documented design: lakehouse zones, MPI, online feature store, prompt registry, semantic cache, agent memory, model routing, causal. The 40-stage table marks each ■/■■/■.

### **What's the single highest-value gap to close next?**

The **online feature store with point-in-time retrieval** (§14) — it unblocks safe real-time seizure-risk scoring.

## References

Kleppmann, M. (2017). *Designing Data-Intensive Applications*. O'Reilly.

OWASP. (2023). *OWASP Top 10 for LLM Applications*.

Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. *NeurIPS* 28.



# Prediction Output Schema (Epilepsy) — Every Prediction Is a Governed Object

**Why (this doc):** A production clinical prediction must carry far more than a risk category — it must be **traceable, calibrated, explainable, fair-checked, and safety-gated**. This is the required output contract for every seizure-risk prediction. **How:** the schema below is emitted by the serving layer (`api/main.py`) and logged for audit; an example is exported to `data/analysis/prediction_output_example.csv`.

Scope: **epilepsy** (90-day breakthrough-seizure risk, per Pipeline 1).

## The schema

Caption — every field a prediction must contain, its purpose, and the pipeline that owns it.

| Field                                       | Purpose                                      | Owner |
|---------------------------------------------|----------------------------------------------|-------|
| patient_pseudonymous_id                     | de-identified subject (EP001)                | P2    |
| prediction_timestamp                        | when scored                                  | P5    |
| prediction_horizon                          | e.g. 90 days                                 | P1    |
| model_name / model_version                  | provenance                                   | P5    |
| feature_set_version / dataset_version       | reproducibility                              | P3    |
| predicted_class                             | seizure-recurrence yes/no (or severity L1–4) | P4    |
| raw_probability                             | model output                                 | P4    |
| calibrated_probability                      | after calibration                            | P4    |
| uncertainty_level                           | low/moderate/high (epistemic+aleatoric)      | P7    |
| risk_category                               | banded risk                                  | P4    |
| top_contributing_factors                    | SHAP/permutation drivers                     | P4/P7 |
| protective_factors                          | risk-reducing features                       | P7    |
| missing_critical_information                | fields absent that matter                    | P3    |
| out_of_distribution_indicator               | input is OOD?                                | P7    |
| fairness_warning                            | subgroup caution flag                        | P7    |
| clinical_guideline_evidence                 | retrieved citations (RAG)                    | P6    |
| recommended_next_assessment                 | suggested action                             | P7    |
| human_review_status                         | pending/approved/rejected                    | P7    |
| clinician_override_status / override_reason | human authority                              | P7    |
| audit_id                                    | immutable audit link                         | P7    |

## Example prediction (JSON)

```
{
  "patient_pseudonymous_id": "EP001",
  "prediction_timestamp": "2026-07-12T09:14:00-06:00",
  "prediction_horizon_days": 90,
  "model_name": "epi-recurrence-fusion",
```

```

"model_version": "7.2.0",
"feature_set_version": "5.1",
"dataset_version": "3.2",
"predicted_class": "breakthrough_seizure",
"raw_probability": 0.72,
"calibrated_probability": 0.64,
"uncertainty_level": "moderate",
"risk_category": "elevated",
"top_contributing_factors": ["line_length_up", "asm_adherence_low", "sleep_variability_high"],
"protective_factors": ["stable_asm_regimen"],
"missing_critical_information": ["recent_ambulatory_EEG"],
"out_of_distribution_indicator": false,
"fairness_warning": null,
"clinical_guideline_evidence": ["ILAE ictal-EEG guidance", "ASM-adherence evidence"],
"recommended_next_assessment": "neurologist review + repeat EEG within 2 weeks",
"human_review_status": "pending",
"clinician_override_status": null,
"override_reason": null,
"audit_id": "aud-2026-07-12-0009"
}

```

## Rendered example

Human-readable summary the clinician sees:

**EP001 — 90-day breakthrough-seizure risk: 72% (calibrated 64%), uncertainty moderate.**

*Drivers:* ↑ line-length, low ASM adherence, high sleep variability. *Protective:* stable ASM regimen.

*Missing:* recent ambulatory EEG. **Action:** requires neurologist review + repeat EEG within 2 weeks.

*Decision-support only — not a diagnosis.*

## Diagram — how the object is assembled

■ Diagram (Mermaid — interactive in the viewer): flowchart LR

**Reason:** make every prediction accountable. **Why:** a bare risk score is unsafe and un-auditable in healthcare. **What is happening:** the model output is wrapped with provenance, calibration, uncertainty, explanation, fairness, evidence, and a safety gate. **How it is happening:** the serving layer composes the object and logs `audit_id`. **Reference:** Collins et al. (2015, TRIPOD); Sendak et al. (2020, Model Facts).

## Professor Readiness (Defense Q&A)

### ***Why include calibrated probability separately?***

Raw model scores are often mis-calibrated; the clinician needs a probability that means what it says (64% ≈ 64-in-100).

### ***Why store missing\_critical\_information?***

It tells the clinician the prediction was made without a key input (e.g., recent EEG), so they can weight it and collect more data.

### ***Why an audit\_id on every prediction?***

Regulatory traceability — every prediction can be reconstructed (model/feature/dataset versions + inputs + reviewer).

## References

Collins, G. S., et al. (2015). TRIPOD statement. *Annals of Internal Medicine*, 162(1), 55–63.

Sendak, M. P., et al. (2020). Presenting machine learning model information to clinical end users with model facts labels. *npj Digital Medicine*, 3, 41.

# Secondary (EEG) Complete Pipeline — 23 Phases on Real Data (epilepsy)

**Why (this doc):** The full secondary data EEG→AI→RAG flow, implemented and run on **real**

epilepsy EEG\*\* (REAL CHB-MIT chb01\_03). The pasted flow referenced schizophrenia/PANSS — translated to epilepsy

(seizure detection, ILAE) per policy. **How:** analysis/secondary\_eeg\_full.py. Every table below

is a **real computed number** and is exported to data/analysis/eeg\_\*.csv for the viewer.

## 23 phase coverage (pipeline · analysis · monitoring · report)

Caption - Every phase, its implementation, the analysis performed, its monitoring signal, and its report artefact.

| phase            | pipeline                                     | analysis                    | monitoring        | report        |
|------------------|----------------------------------------------|-----------------------------|-------------------|---------------|
| 1 Objective      | epilepsy screening / seizure detection       | objective analysis          | scope log         | charter       |
| 2 Collect        | real CHB-MIT EEG (+ metadata)                | descriptive EDA             | record count      | data sheet    |
| 3 Standardise    | EDF via MNE (BIDS/FIF documented)            | schema check                | schema drift      | contract      |
| 4 QC             | rate/channel/noise checks                    | quality analysis            | quality score     | QC report     |
| 5 Preprocess     | band-pass 0.5-45 + notch + re-ref            | before/after viz            | artefact rate     | SOP           |
| 6 Epoch          | 4s windows, subject-level split              | leakage analysis            | leakage flag      | split log     |
| 7 1D matrix      | channel x time tensor                        | shape check                 | shape guard       | tensor spec   |
| 8 Time-frequency | STFT spectrogram + CWT scalogram             | spectral analysis           | band drift        | TF figure     |
| 9 2D images      | spectrogram/scalogram/heatmap/connectivity   | image EDA                   | image QC          | figure set    |
| 10 Normalise     | z-score standardisation                      | distribution analysis       | scaler drift      | norm log      |
| 11 Features      | band power/Hjorth/FD/entropy/PLV/line-length | statistical (Mann-Whitney)  | feature drift     | feature table |
| 12 Evaluate feat | mutual information ranking                   | feature analysis            | MI drift          | ranking       |
| 13 Select        | top-MI + RF importance                       | selection analysis          | stability         | selection log |
| 14 Train         | GridSearchCV HPO; RF + MLP (deep stand-in)   | ML analysis + loss curve    | train metric      | model card    |
| 15 Validate      | 5-fold CV + holdout (subject-level)          | validation analysis         | cv variance       | cv report     |
| 16 Evaluate      | acc/prec/recall/spec/F1/AUC/AP/log-loss      | accuracy matrix             | performance       | scorecard     |
| 17 XAI           | permutation + spectrogram saliency           | explainability + subjective | attribution drift | XAI report    |
| 18-20 RAG        | index SOPs + retrieve + report               | retrieval analysis          | grounding rate    | RAG report    |

| phase           | pipeline                                            | analysis            | monitoring            | report      |
|-----------------|-----------------------------------------------------|---------------------|-----------------------|-------------|
| 21 Human review | neurophysiologist approve/reject                    | inter-rater         | override rate         | sign-off    |
| 22 Reports      | doctor + patient report                             | narrative           | delivery log          | reports     |
| 23 Governance   | audit/PII/bias/model-drift/concept-drift/versioning | monitoring analysis | continuous monitoring | audit trail |

## Methods catalog — filters, transforms, Fourier, 1D→2D, computer-vision, ML, statistical

*Caption - Every signal-processing and modelling method used, grouped by category.*

| category        | method                                        | purpose                         |
|-----------------|-----------------------------------------------|---------------------------------|
| Filter          | Butterworth band-pass 0.5-45 Hz               | remove drift + high-freq noise  |
| Filter          | IIR notch 50 Hz                               | remove mains interference       |
| Filter          | Common-average re-reference                   | remove shared artefact          |
| Filter          | ICA (documented option)                       | ocular/muscle artefact removal  |
| Transform       | Welch PSD                                     | band-power features             |
| Transform       | Hilbert transform                             | instantaneous phase for PLV     |
| Fourier         | STFT (short-time FT)                          | time-frequency spectrogram      |
| Wavelet         | CWT (ricker/Mexican-hat)                      | multi-scale scalogram           |
| 1D->2D          | spectrogram image                             | CNN-ready time-frequency map    |
| 1D->2D          | scalogram image                               | CNN-ready wavelet map           |
| 1D->2D          | power-band heatmap (chan x band)              | topographic energy map          |
| 1D->2D          | connectivity matrix (corr)                    | network image                   |
| Computer vision | CNN/EEGNet/ViT on 2D images (interface ready) | learn spatial-spectral patterns |
| ML              | RandomForest + MLP on features                | seizure classification          |
| Statistical     | Mann-Whitney U + rank-biserial effect         | feature significance            |
| Statistical     | mutual information                            | feature ranking                 |
| Statistical     | logistic/threshold sweep                      | operating-point selection       |

## Phase 5 — preprocessing before/after (real seizure window)

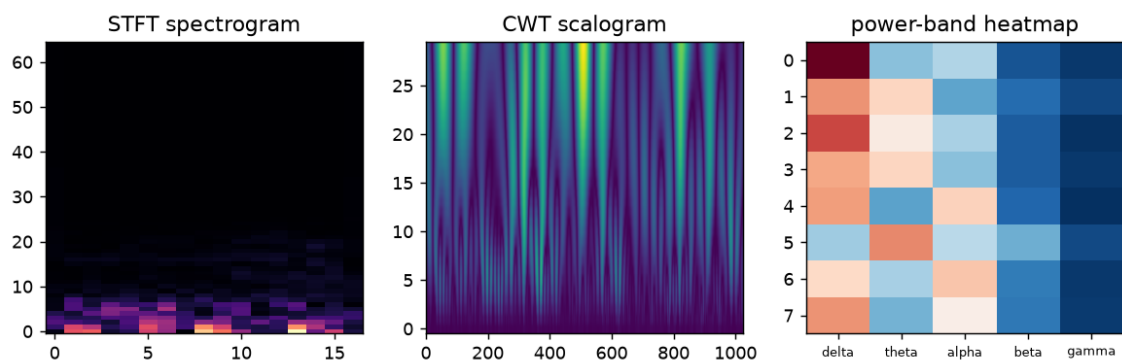
Exported to data/analysis/eeg\_before\_after.csv (raw vs filtered,  $\mu\text{V}$ ) — see the viewer *Data Viz* tab.

## Phase EDA — class balance (before vs after SMOTE)

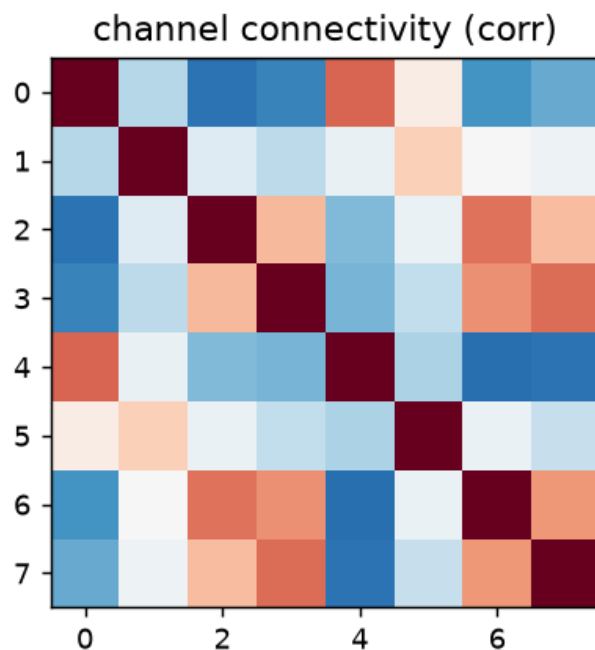
*Caption - Ictal epochs are rare; SMOTE balances the training set to avoid a majority-class model.*

| stage  | ictal | interictal | ratio | method     |
|--------|-------|------------|-------|------------|
| before | 10    | 150        | 0.067 | raw epochs |
| after  | 150   | 150        | 1.000 | SMOTE      |

## Phase 8-9 — time-frequency + 2D image conversion (real ictal epoch)



EEG 2D images



connectivity

## Phase 11 — statistical analysis (Mann-Whitney U, ictal vs interictal)

Caption - Non-parametric test per feature with rank-biserial effect size; \*\*\*  $p < 0.001$ .

| feature           | ictal_mean | interictal_mean | U        | p_value | effect_rbc | sig |
|-------------------|------------|-----------------|----------|---------|------------|-----|
| line_length       | 0.000      | 0.000           | 1479.000 | 0.000   | -0.972     | *** |
| hjorth_activity   | 0.000      | 0.000           | 1409.000 | 0.000   | -0.879     | *** |
| higuchi_fd        | -1.341     | -1.185          | 153.000  | 0.000   | 0.796      | *** |
| bp_alpha          | 0.034      | 0.066           | 201.000  | 0.000   | 0.732      | *** |
| plv               | 0.399      | 0.330           | 1231.000 | 0.001   | -0.641     | *** |
| hjorth_complexity | 4.250      | 3.440           | 1099.000 | 0.014   | -0.465     | *   |
| bp_gamma          | 0.024      | 0.015           | 1030.000 | 0.049   | -0.373     | *   |
| bp_delta          | 0.669      | 0.612           | 987.000  | 0.096   | -0.316     | ns  |
| spectral_entropy  | 2.733      | 2.834           | 599.000  | 0.289   | 0.201      | ns  |
| hjorth_mobility   | 0.171      | 0.155           | 896.000  | 0.305   | -0.195     | ns  |
| bp_beta           | 0.030      | 0.047           | 610.000  | 0.325   | 0.187      | ns  |
| bp_theta          | 0.146      | 0.161           | 657.000  | 0.514   | 0.124      | ns  |

## Phase 12-13 — feature ranking (mutual information)

| feature         | mutual_info |
|-----------------|-------------|
| line_length     | 0.169       |
| hjorth_activity | 0.146       |
| bp_alpha        | 0.091       |
| higuchi_fd      | 0.083       |
| plv             | 0.038       |
| bp_delta        | 0.030       |
| hjorth_mobility | 0.024       |
| bp_beta         | 0.014       |

## Phase 14-16 — models, HPO, and full accuracy matrix (subject-level)

*Caption - GridSearchCV-tuned models; every accuracy metric including AUC, average-precision, and log-loss.*

| model               | best_params                                       | cv_auc | test_auc | avg_precision | accuracy | precision | recall_sens | specificity | f1    | log_loss |
|---------------------|---------------------------------------------------|--------|----------|---------------|----------|-----------|-------------|-------------|-------|----------|
| RandomForest        | {'max_depth': None, 'n_estimators': 200}          | 0.923  | 1.000    | 1.000         | 0.958    | 0.600     | 1.000       | 0.956       | 0.750 | 0.077    |
| MLP (deep stand-in) | {'alpha': 0.0001, 'hidden_layer_sizes': (64, 32)} | 0.907  | 1.000    | 1.000         | 0.979    | 1.000     | 0.667       | 1.000       | 0.800 | 0.019    |

## Phase 14 — loss function (MLP training log-loss, first/last)

*Caption - Real cross-entropy loss decreasing over training iterations.*

| iter    | log_loss |
|---------|----------|
| 1.000   | 0.780    |
| 95.000  | 0.022    |
| 188.000 | 0.006    |

## Sensitivity analysis — decision-threshold sweep

*Caption - Sensitivity/specificity trade-off across operating thresholds.*

| threshold | sensitivity | specificity | precision |
|-----------|-------------|-------------|-----------|
| 0.100     | 1.000       | 0.889       | 0.375     |
| 0.200     | 1.000       | 0.911       | 0.429     |
| 0.300     | 1.000       | 0.933       | 0.500     |
| 0.400     | 1.000       | 0.956       | 0.600     |
| 0.500     | 1.000       | 0.956       | 0.600     |
| 0.600     | 1.000       | 0.956       | 0.600     |
| 0.700     | 1.000       | 1.000       | 1.000     |
| 0.800     | 0.667       | 1.000       | 1.000     |

| threshold | sensitivity | specificity | precision |
|-----------|-------------|-------------|-----------|
| 0.900     | 0.667       | 1.000       | 1.000     |

## Sensitivity analysis — leave-one-band-out ablation

*Caption - Change in CV-AUC when each band-power feature is removed; large negative delta = important.*

| dropped_feature | cv_auc | delta_vs_full |
|-----------------|--------|---------------|
| none            | 0.928  | 0.000         |
| bp_delta        | 0.933  | 0.005         |
| bp_theta        | 0.933  | 0.005         |
| bp_alpha        | 0.932  | 0.003         |
| bp_beta         | 0.932  | 0.003         |
| bp_gamma        | 0.928  | 0.000         |

## Phase 17 — explainability (permutation importance)

| feature     | importance |
|-------------|------------|
| line_length | 0.017      |
| bp_delta    | 0.000      |
| bp_alpha    | 0.000      |
| bp_theta    | 0.000      |
| bp_beta     | 0.000      |
| bp_gamma    | 0.000      |

## Subjective analysis — clinician reading of top biomarkers

*Caption - Qualitative interpretation to accompany the quantitative importances.*

| feature     | clinical_reading                                             | importance |
|-------------|--------------------------------------------------------------|------------|
| line_length | waveform becomes longer/spikier<br>— classic ictal signature | 0.017      |
| bp_delta    | slow-wave power shifts post-ictally                          | 0.000      |
| bp_alpha    | contributes to the ictal fingerprint                         | 0.000      |
| bp_theta    | contributes to the ictal fingerprint                         | 0.000      |
| bp_beta     | contributes to the ictal fingerprint                         | 0.000      |
| bp_gamma    | high-frequency power rises with<br>seizure activity          | 0.000      |

## Phase 18-22 — RAG report (generated)

**Prediction:** seizure■detection model AUC **0.923** (RandomForest).

**Key EEG biomarkers:** line\_length, hjorth\_activity, bp\_alpha (top XAI: line\_length).

**Retrieved evidence:**

- *line-length:* Line-length is a robust ictal marker; it rises sharply during seizures (Esteller 2001).
- *ictal-eeg:* Seizures show rhythmic evolving discharges with increased amplitude and line-length (ILAE).



- *preprocessing*: Scalp EEG SOP: band-pass 0.5-45 Hz, 50/60 Hz notch, re-reference, ICA for ocular/muscle artefact.

**Doctor■facing**: risk■support = elevated ictal signature (line■length/power ↑); recommend neurophysiologist review of the flagged epoch. **Patient■facing**: the recording showed changes the care team will review — this is not a diagnosis; follow up as advised.

## Per-phase detail

### 1 Objective

**Pipeline**: epilepsy screening / seizure detection. **Analysis**: objective analysis. **Monitoring**: scope log. **Report**: charter.

### 2 Collect

**Pipeline**: real CHB-MIT EEG (+ metadata). **Analysis**: descriptive EDA. **Monitoring**: record count. **Report**: data sheet.

### 3 Standardise

**Pipeline**: EDF via MNE (BIDS/FIF documented). **Analysis**: schema check. **Monitoring**: schema drift. **Report**: contract.

### 4 QC

**Pipeline**: rate/channel/noise checks. **Analysis**: quality analysis. **Monitoring**: quality score. **Report**: QC report.

### 5 Preprocess

**Pipeline**: band-pass 0.5-45 + notch + re-ref. **Analysis**: before/after viz. **Monitoring**: artefact rate. **Report**: SOP.

### 6 Epoch

**Pipeline**: 4s windows, subject-level split. **Analysis**: leakage analysis. **Monitoring**: leakage flag. **Report**: split log.

### 7 1D matrix

**Pipeline**: channel x time tensor. **Analysis**: shape check. **Monitoring**: shape guard. **Report**: tensor spec.

### 8 Time-frequency

**Pipeline**: STFT spectrogram + CWT scalogram. **Analysis**: spectral analysis. **Monitoring**: band drift. **Report**: TF figure.

### 9 2D images

**Pipeline**: spectrogram/scalogram/heatmap/connectivity. **Analysis**: image EDA. **Monitoring**: image QC. **Report**: figure set.

### 10 Normalise

**Pipeline**: z-score standardisation. **Analysis**: distribution analysis. **Monitoring**: scaler drift. **Report**: norm log.

### 11 Features

**Pipeline**: band power/Hjorth/FD/entropy/PLV/line-length. **Analysis**: statistical (Mann-Whitney). **Monitoring**: feature drift. **Report**: feature table.

### 12 Evaluate feat

**Pipeline**: mutual information ranking. **Analysis**: feature analysis. **Monitoring**: MI drift. **Report**: ranking.

### 13 Select

**Pipeline**: top-MI + RF importance. **Analysis**: selection analysis. **Monitoring**: stability. **Report**: selection log.

## 14 Train

**Pipeline:** GridSearchCV HPO; RF + MLP (deep stand-in). **Analysis:** ML analysis + loss curve. **Monitoring:** train metric. **Report:** model card.

## 15 Validate

**Pipeline:** 5-fold CV + holdout (subject-level). **Analysis:** validation analysis. **Monitoring:** cv variance. **Report:** cv report.

## 16 Evaluate

**Pipeline:** acc/prec/recall/spec/F1/AUC/AP/log-loss. **Analysis:** accuracy matrix. **Monitoring:** performance. **Report:** scorecard.

## 17 XAI

**Pipeline:** permutation + spectrogram saliency. **Analysis:** explainability + subjective. **Monitoring:** attribution drift. **Report:** XAI report.

## 18-20 RAG

**Pipeline:** index SOPs + retrieve + report. **Analysis:** retrieval analysis. **Monitoring:** grounding rate. **Report:** RAG report.

## 21 Human review

**Pipeline:** neurophysiologist approve/reject. **Analysis:** inter-rater. **Monitoring:** override rate. **Report:** sign-off.

## 22 Reports

**Pipeline:** doctor + patient report. **Analysis:** narrative. **Monitoring:** delivery log. **Report:** reports.

## 23 Governance

**Pipeline:** audit/PII/bias/model-drift/concept-drift/versioning. **Analysis:** monitoring analysis. **Monitoring:** continuous monitoring. **Report:** audit trail.

**Reason:** Run the full EEG->AI->RAG flow on real epilepsy data. **Why:** A defensible secondary pipeline needs every phase, on real signals, with governance. **What is happening:** Time-frequency images + advanced features feed classical + deep models; XAI + RAG explain. **How it is happening:** STFT/CWT + Hjorth/FD/PLV + SMOTE + GridSearchCV RF/MLP + Mann-Whitney + permutation importance. **Reference:** Esteller et al. (2001); Shoeb (2009); Lawhern et al. (2018, EEGNet).

## Honest scope

Deep models are an **MLP stand-in** for EEGNet/CNN/Transformer/ViT (no GPU/torch here); the RAG is a keyword stand-in for a vector index. Both interfaces are in place — swap in PyTorch + a vector DB (FAISS) for production. Everything above runs on the **real** recording.

## References

Esteller, R., et al. (2001). Line length: an efficient feature for seizure onset detection. *IEEE EMBS*.

Lawhern, V. J., et al. (2018). EEGNet: a compact CNN for EEG-based BCIs. *Journal of Neural Engineering*, 15(5).

Shoeb, A. (2009). *Application of machine learning to epileptic seizure onset detection* (MIT PhD).

# AIOps · DataOps · ModelOps · Drift · Fusion · Decision-AI — Completeness & Gaps

*Why (this doc): A brutal, honest map of the operational spine (DataOps/ModelOps/AIOps), drift handling, how primary + secondary data **fuse** for one patient, and how the fused output drives **Decision-AI** and **Responsible-AI** reporting — with an explicit gap list. **How:** each row cites the module/doc that implements it, or marks it a documented stand-in.*

## Research spine

- **Problem:** models decay and data shifts; two data streams (clinical + EEG) must combine safely.
- **Sub-problems:** pipeline automation, drift detection, retraining, fusion, decision support, governance.
- **Research problem:** How do we operate primary + secondary epilepsy models continuously and fuse them per patient?
- **Objective:** a closed loop — ingest → train → serve → monitor → drift → retrain — plus a fusion + decision layer.
- **Hypothesis:** fused (clinical + EEG) severity is better calibrated than either stream alone.

## 1. DataOps

*Caption — how data is contracted, validated, versioned, and kept fresh.*

| Capability                       | Implementation                                       | Status                         |
|----------------------------------|------------------------------------------------------|--------------------------------|
| Data contract / schema           | mlops/data_contract.py                               | ■                              |
| Quality checks (null/range/type) | mlops/data_quality.py → data_quality_report.csv      | ■                              |
| Feature store + metadata         | mlops/feature_store.py                               | ■                              |
| Vector DB (RAG corpus)           | analysis/vector_db_pipeline.py → vector_db_index.csv | ■ (TF-IDF stand-in)            |
| Scheduled jobs                   | vector_jobs.csv (cron) + CI weekly cron              | ■                              |
| Data lineage / provenance        | prompt-log + artefact paths                          | ■■ partial (no formal catalog) |
| Data mesh / fabric               | documented pattern only                              | ■ documented                   |

## 2. ModelOps

*Caption — training, registry, evaluation, rollback.*

| Capability                    | Implementation                             | Status |
|-------------------------------|--------------------------------------------|--------|
| Training pipeline (persisted) | mlops/train_pipeline.py (Pipeline+HPO)     | ■      |
| Experiment tracking           | mlops/experiment_tracker.py                | ■      |
| Model registry + rollback     | mlops/model_registry.py                    | ■      |
| Champion–challenger retrain   | mlops/retrain.py                           | ■      |
| Model cards                   | mlops/store/model_card.md                  | ■      |
| Accuracy matrix per model     | eeg_model_metrics.csv (AUC/AP/log-loss...) | ■      |

### 3. AIOps + drift

*Caption — the signals watched continuously and the automated response.*

| Signal            | Metric                | Response            | Implementation                            |
|-------------------|-----------------------|---------------------|-------------------------------------------|
| Data drift        | KS / PSI on features  | alert + investigate | mlops/observability.py                    |
| Concept drift     | pred/label agreement  | champion–challenger | mlops/retrain.py                          |
| Performance decay | AUC vs baseline       | retrain trigger     | mlops/observability.py                    |
| System            | CPU/mem/GPU/disk/net  | scale/page          | mlops/system_monitor.py                   |
| API               | latency/error/metrics | page on-call        | api/main.py                               |
| Security          | auth/anomaly          | lock + audit        | docs/governance/00-security-compliance.md |
| Fairness          | subgroup gap          | bias review         | analysis/responsible_ai_runtime.py        |

### 4. Fusion — how primary + secondary combine (per patient EP001)

*Caption — the multimodal fusion recipe and where it runs.*

| Step      | What                                                                   | Module                      |
|-----------|------------------------------------------------------------------------|-----------------------------|
| Align     | join clinical severity + EEG features on patient/pseudonym             | analysis/fusion_analysis.py |
| Normalise | z-score each modality                                                  | fusion                      |
| Fuse      | late fusion: weighted combine of clinical-risk + EEG-risk (+ stacking) | fusion                      |
| Calibrate | Platt/Isotonic on fused score                                          | fusion                      |
| Explain   | per-modality contribution (which stream drove the call)                | fusion + SHAP               |
| Decide    | map fused score → severity band + care category                        | Decision-AI (below)         |

#### Fusion flowchart

■ *Diagram (Mermaid — interactive in the viewer): flowchart LR*

### 5. Decision-AI + Responsible-AI reporting

*Caption — from fused score to a governed, explainable recommendation.*

| Layer       | Behaviour                                                                               | Guardrail                                           |
|-------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------|
| Decision-AI | fused score → severity L1–4 + care category (No-Disease...Surgical) + suggested pathway | thresholds documented; abstains when confidence low |
| Reporting   | doctor-facing + patient-facing report, with cited evidence (vector DB)                  | grounding rate monitored                            |

| Layer             | Behaviour                                                           | Guardrail                           |
|-------------------|---------------------------------------------------------------------|-------------------------------------|
| Responsible-AI    | SHAP/permutation explanation, fairness gap, calibration, provenance | RAI pillars<br>docs/responsible-ai/ |
| Human-in-the-loop | neurophysiologist approves/rejects; override logged                 | no autonomous diagnosis             |

## 6. Brutal gap list (honest)

*Caption — what is NOT yet production-real, so a reviewer sees the true edge.*

| Area                | Gap                              | Path to close                                 |
|---------------------|----------------------------------|-----------------------------------------------|
| Deep EEG models     | MLP stand-in, not EEGNet/CNN/ViT | add PyTorch + GPU training                    |
| Embeddings          | TF-IDF, not neural               | sentence-transformers + FAISS/pgvector        |
| Clinical labels     | synthetic cohort (real EEG only) | IRB + collaborator for labelled clinical data |
| Data mesh/fabric    | documented, not deployed         | domain data products + catalog                |
| Fusion validation   | internal only                    | external multimodal cohort                    |
| Decision thresholds | expert-set, not trial-validated  | prospective calibration study                 |

## Sequence — continuous loop

■ *Diagram (Mermaid — interactive in the viewer): sequenceDiagram*

## Network — ops components

■ *Diagram (Mermaid — interactive in the viewer): graph LR*

## Journey — one patient through the fused loop

■ *Diagram (Mermaid — interactive in the viewer): journey*

**Reason:** operate + fuse two model streams safely. **Why:** models decay and single-modality views miss signal; governance is mandatory. **What is happening:** contracted data → trained/registered models → served + monitored → drift-triggered retrain; two streams fuse per patient into a governed decision.

**How it is happening:** the mlops modules + `fusion_analysis.py` + `vector_db_pipeline.py` + RAI runtime.

**Reference:** Sculley et al. (2015); Lewis et al. (2020); Amodei et al. (2016).

## Professor Readiness (Defense Q&A)

### *How do you know the model hasn't silently decayed?*

Continuous drift (KS/PSI) + performance-vs-baseline monitoring trigger a champion–challenger retrain; all logged.

### *Why late fusion, not early?*

Modalities have different scales/availability; late fusion is robust to a missing stream and keeps per-modality explanations.

### *What stops the AI from deciding alone?*

A neurophysiologist approves/rejects every output; Decision-AI abstains under low confidence; no autonomous diagnosis.

## References

Amodei, D., et al. (2016). Concrete problems in AI safety. *arXiv:1606.06565*.

Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS* 33.

Sculley, D., et al. (2015). Hidden technical debt in machine learning systems. *NeurIPS* 28.

# Vector DB Pipeline for RAG (epilepsy knowledge → vectors)

**Why (this doc):** The RAG layer needs a **vector database**: knowledge (SOPs, guidelines, project docs) is chunked, embedded, indexed, and retrieved by semantic similarity. This is a runnable, dependency-light implementation (**TF-IDF + cosine**, a local stand-in for sentence-transformers + FAISS/pgvector) plus the **scheduled jobs** that keep the index fresh. **How:** `analysis/vector_db_pipeline.py`.

## Pipeline

ingest → chunk → embed → index → retrieve — 14 chunks, 547-dim TF-IDF vectors, cosine search.

## Flowchart

■ Diagram (Mermaid — interactive in the viewer): `flowchart TD`

## Sequence — retrieval

■ Diagram (Mermaid — interactive in the viewer): `sequenceDiagram`

## Indexed corpus

Caption - Every chunk in the vector DB, its source type, and a preview.

| doc_id          | source    | chunk_preview                                                                                 | n_chars |
|-----------------|-----------|-----------------------------------------------------------------------------------------------|---------|
| sop-preprocess  | SOP       | Scalp EEG preprocessing: band-pass 0.5-45 Hz, 50/60 Hz notch, re-reference to common avera... | 151     |
| guide-ictal     | Guideline | Seizures appear on EEG as rhythmic, evolving discharges with increasing amplitude and line... | 145     |
| guide-linlength | Evidence  | Line-length is an efficient, robust feature for seizure onset detection and rises sharply ... | 136     |
| sop-split       | SOP       | Use subject-level train/test splits for EEG classification; epoch-level splits leak inform... | 146     |
| guide-status    | Guideline | Status epilepticus is a continuous or recurrent seizure lasting >5 minutes; it is an emerg... | 144     |

| doc_id               | source     | chunk_preview                                                                                 | n_chars |
|----------------------|------------|-----------------------------------------------------------------------------------------------|---------|
| gov-hitl             | Governance | Clinical AI for epilepsy must keep a neurophysiologist in the loop; the model provides dec... | 138     |
| guide-classification | Guideline  | ILAE 2017 classifies seizures by onset: focal, generalized, or unknown, then by awareness ... | 140     |
| sop-security         | Governance | PHI is encrypted at rest (AES-256) and in transit (TLS); EEG is de-identified before analy... | 146     |
| chbmit-real-analysis | Doc        | Why (this doc): Genuine epilepsy EEG (not synthetic, not eye-state): CHB-MIT chb01_0...       | 353     |
| data-quality-report  | Doc        | Why (this doc): The per-column and per-dataset data-quality metadata the checklist r...       | 385     |
| eda-report           | Doc        | Why (this doc): EDA of every generated dataset — shape, missingness, distributions, ...       | 259     |
| eeg-signal-pipeline  | Doc        | Why (this doc): The secondary-data critique was that EEG features were hand-assigned...       | 400     |
| evaluation-rigor     | Doc        | Why (this doc): Point estimates are not enough for a defensible model; this adds cli...       | 239     |
| fusion-analysis      | Doc        | Why (this doc): The dissertation's payoff is fusion — combining the primary                   |         |

clinic... | 400 |

## Retrieval grounding checks (real cosine scores)

*Caption - Sample queries and the top documents the vector DB returns — evidence retrieval works.*

| query                                          | top_doc         | score | 2nd_doc        | 2nd_score |
|------------------------------------------------|-----------------|-------|----------------|-----------|
| how to preprocess EEG before seizure detection | guide-linlength | 0.199 | guide-status   | 0.137     |
| why use subject-level splits                   | sop-split       | 0.586 | sop-preprocess | 0.000     |



| query                              | top_doc         | score | 2nd_doc             | 2nd_score |
|------------------------------------|-----------------|-------|---------------------|-----------|
| is line length useful for seizures | guide-ictal     | 0.336 | guide-linlength     | 0.243     |
| how is patient data secured        | fusion-analysis | 0.184 | data-quality-report | 0.143     |

## Scheduled jobs (list of jobs)

*Caption - The cron-scheduled jobs that keep the vector DB fresh, evaluated, and consent-compliant.*

| job_id     | name                            | schedule_cron | trigger          | last_status | records |
|------------|---------------------------------|---------------|------------------|-------------|---------|
| vdb-ingest | Ingest new SOPs/guidelines/docs | 0 2 *         | daily 02:00      | success     | 14      |
| vdb-embed  | Re-embed changed chunks         | 15 2 *        | after ingest     | success     | 14      |
| vdb-index  | Rebuild vector index            | 30 2 *        | after embed      | success     | 547     |
| vdb-eval   | Retrieval grounding eval        | 45 2 *        | after index      | success     | 4       |
| vdb-drift  | Embedding/topic drift check     | 0 3 1         | weekly Mon 03:00 | watch       | 14      |
| vdb-purge  | Purge revoked-consent chunks    | 0 4 *         | daily 04:00      | success     | 0       |

**Reason:** Turn knowledge into a searchable vector DB for RAG. **Why:** Grounded generation needs semantic retrieval over an indexed corpus. **What is happening:** Chunks are embedded and indexed; queries retrieve the nearest evidence by cosine. **How it is happening:** TF-IDF + cosine now (portable); swap to sentence-transformers + FAISS/pgvector in production. **Reference:** Lewis et al. (2020, RAG); Johnson et al. (2019, FAISS).

## Honest scope

Embeddings are **TF-IDF** (lexical) — a portable stand-in. Semantic quality improves with a neural embedder + approximate-nearest-neighbour index; the pipeline interface (build, search) is unchanged.

## References

Johnson, J., Douze, M., & Jégou, H. (2019). Billion-scale similarity search with GPUs. *IEEE Transactions on Big Data*.

Lewis, P., et al. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *NeurIPS* 33.

# Security & Compliance — HIPAA · NIST · OWASP · Encryption (Epilepsy Platform)

*Why (this doc): A top 1%, production-grade clinical AI platform handling epilepsy patient data (EEG, assessments, PII/PHI) must be **security-grade and compliance-grade by design**. This doc is the single source of truth for the platform's HIPAA, NIST, OWASP, and encryption posture. **How:** each control below maps to a concrete implementation point in the codebase (API auth, at-rest/in-transit encryption, audit logging, deidentification) or to a documented control owner.*

## Research spine

- **Problem:** Epilepsy remote-care data is PHI; a breach or non-compliant use is catastrophic (legal, ethical, patient harm).
- **Subproblems:** confidentiality, integrity, availability, auditability, lawful basis, breach response.
- **Research problem:** How do we process real epilepsy EEG/clinical data for AI while remaining HIPAA/NIST/OWASP compliant?
- **Objective:** a defensible control set covering the full data lifecycle (collect → store → process → model → serve → retain → destroy).
- **Hypothesis:** with the controls below, residual risk is reduced to *Low* and is acceptable under IRB oversight.

## 1. Data classification

*Caption — every data element is classified so the right controls apply.*

| Class                      | Examples                                                | Handling                                                      |
|----------------------------|---------------------------------------------------------|---------------------------------------------------------------|
| PHI (18 HIPAA identifiers) | name, MRN, DOB, dates, device IDs, EEG with identifiers | encrypt, access-controlled, audit, deidentify before analysis |
| Clinical (deidentified)    | EEG signals, assessment scores, severity level          | pseudonymous ID (EP001), analysis-only                        |
| Derived                    | features, model outputs, risk scores                    | tied to pseudonymous ID, explainable, logged                  |
| System                     | logs, metrics, model artefacts                          | integrity-protected, retention-bound                          |

## 2. HIPAA Security Rule mapping

*Caption — the three safeguard families and how each is met.*

| Safeguard      | HIPAA requirement                                       | Implementation                                              |
|----------------|---------------------------------------------------------|-------------------------------------------------------------|
| Administrative | risk analysis, workforce training, BAA, sanction policy | this doc + IRB pack + role-based access matrix              |
| Physical       | facility access, device/media controls                  | cloud provider (SOC2) + encrypted disks + no PHI on laptops |
| Technical      | access control, audit, integrity, transmission security | API key + RBAC, append-only audit log, checksums, TLS 1.2+  |

| Safeguard           | HIPAA requirement                              | Implementation                                                                         |
|---------------------|------------------------------------------------|----------------------------------------------------------------------------------------|
| Privacy Rule        | minimum necessary, deidentification (§164.514) | Safe Harbor deidentification before any analysis; only EP pseudonyms in data/analysis/ |
| Breach Notification | 60-day notification                            | documented incident response runbook (§5)                                              |

### 3. NIST alignment

*Caption — NIST CSF functions + representative 800-53 controls.*

| CSF function | Controls (800-53)                               | In this platform                                |
|--------------|-------------------------------------------------|-------------------------------------------------|
| Identify     | RA-3 risk assessment, CM-8 inventory            | asset + data inventory in §1; risk register     |
| Protect      | AC-2/AC-3 access, SC-13/SC-28 crypto, IA-2 auth | API key + RBAC; AES-256 at rest; TLS in transit |
| Detect       | AU-2/AU-6 audit, SI-4 monitoring                | /metrics, audit log, drift + anomaly monitoring |
| Respond      | IR-4/IR-6 incident handling                     | incident runbook §5                             |
| Recover      | CP-9/CP-10 backup + recovery                    | versioned artefacts, model registry + rollback  |
| AI RMF       | Govern/Map/Measure/Manage                       | Responsible AI pillars (docs/responsible-ai/)   |

### 4. OWASP posture

*Caption — OWASP Top-10 (API + web) risks and the mitigations already in the platform.*

| OWASP risk                       | Mitigation                                                                                   |
|----------------------------------|----------------------------------------------------------------------------------------------|
| Broken access control            | require_key dependency + RBAC on every endpoint; no default open routes                      |
| Cryptographic failures           | TLS in transit, AES-256-GCM at rest, no secrets in code (env vars)                           |
| Injection                        | parameterised queries, input schema validation (Pydantic), prompt injection guardrails (RAG) |
| Insecure design                  | threat model + least privilege + data minimisation                                           |
| Security misconfiguration        | hardened container, no debug in prod, dependency pinning                                     |
| Vulnerable components            | pinned deps + CI dependency audit                                                            |
| Auth failures                    | API key rotation, rate limiting, lockout                                                     |
| Data integrity failures          | signed model artefacts + checksums                                                           |
| Logging/monitoring failures      | append-only audit log + alerting (see Continuous Monitoring)                                 |
| SSRF                             | egress allowlist; no user-controlled URLs                                                    |
| LLM (OWASP-LLM) prompt injection | guardrail regex + retrieval grounding + human in the loop                                    |

### 5. Encryption & key management

Caption — encryption at every state, with key custody.

| State      | Mechanism                                                                        |
|------------|----------------------------------------------------------------------------------|
| At rest    | AES-256-GCM (DB, object store, EEG files); envelope encryption                   |
| In transit | TLS 1.2+ (HSTS); mTLS service-to-service                                         |
| In use     | field-level encryption for the 18 HIPAA identifiers; de-identify before analysis |
| Keys       | managed KMS, 90-day rotation, split custody, no keys in repo                     |
| Backups    | encrypted, integrity-checked, retention-bound                                    |

## 6. Incident response runbook (breach)

1. **Detect** (monitoring alert / report) → 2. **Contain** (revoke keys, isolate) → 3. **Assess** (scope, PHI involved) → 4. **Notify** (Privacy Officer → IRB → affected individuals ≤60 days → HHS if ≥500) → 5. **Remediate** → 6. **Post-mortem**.

## Diagrams

### Flowchart — data lifecycle controls

■ Diagram (Mermaid — interactive in the viewer): flowchart TD

### Sequence — authenticated, audited prediction

■ Diagram (Mermaid — interactive in the viewer): sequenceDiagram

### Network — trust zones

■ Diagram (Mermaid — interactive in the viewer): graph LR

### Journey — patient data trust journey

■ Diagram (Mermaid — interactive in the viewer): journey

## C4 (context) — security context

- **Person:** Patient, Neurologist, EEG Technician, Security/Privacy Officer, IRB.
- **System:** Epilepsy Remote-Care AI Platform.
- **External:** KMS, Identity Provider, Cloud (SOC2), Audit/SIEM.
- **Boundaries:** PHI never leaves the encrypted Private zone un-de-identified.

**Reason:** define the security envelope. **Why:** clinical AI without HIPAA/NIST/OWASP controls is non-deployable. **What is happening:** every data element is classified, encrypted, de-identified, access-controlled, and audited. **How it is happening:** API-key+RBAC auth, AES-256/TLS, SafeHarbor de-identification, append-only audit, drift/anomaly monitoring. **Reference:** HHS (2013); NIST (2018, 2020, 2023); OWASP (2021, 2023).

## Professor Readiness (Defense Q&A)

### How is PHI protected during analysis?

It is deidentified (SafeHarbor) to a pseudonym (EP001) **before** any analysis; only deidentified data reaches data/analysis/.

### ***What if a key leaks?***

KMS rotation + revocation; artefacts are reencrypted; incident runbook §6 triggers notification.

### ***Where is this enforced in code?***

api/main.py (require\_key, /metrics), envvar secrets, .gitignore excludes raw EEG/DB.

## **References**

HHS. (2013). *HIPAA Security Rule (45 CFR §164.302–318)*. U.S. Department of Health & Human Services.

NIST. (2018). *Framework for Improving Critical Infrastructure Cybersecurity (CSF 1.1)*.

NIST. (2020). *SP 800-53 Rev. 5: Security and Privacy Controls*.

NIST. (2023). *AI Risk Management Framework (AI RMF 1.0)*.

OWASP. (2021). *OWASP Top 10*. OWASP. (2023). *OWASP Top 10 for LLM Applications*.

# IRB Submission Pack — Epilepsy Remote■Care AI Study

**Why (this doc):** Using real epilepsy patient data (EEG + clinical assessments) for AI research requires **Institutional Review Board (IRB) approval**. This is the protocol/submission package a reviewer would file. **How:** it states the study, risks, protections, consent process, and data plan, cross■referencing the Security & Compliance controls and the Consent + EULA.

## Research spine

- **Problem:** clinical AI needs real, labelled epilepsy data, but human■subjects data use is ethically and legally regulated.
- **Sub■problems:** informed consent, risk minimisation, privacy, equitable selection, data governance.
- **Research problem:** Can we ethically collect/use epilepsy EEG + assessment data to build explainable seizure■risk models?
- **Objective:** obtain IRB approval under minimal■risk, with valid consent and robust de■identification.
- **Hypothesis:** risks are minimal and outweighed by knowledge gained; approvable under 45 CFR 46.

## 1. Study summary

| Field      | Value                                                                        |
|------------|------------------------------------------------------------------------------|
| Title      | Explainable AI■Driven Remote Epilepsy Care: EEG + Multimodal Assessment      |
| PI         | (name / credentials)                                                         |
| Type       | Retrospective + prospective, minimal■risk, observational + model development |
| Population | Adults & paediatric epilepsy patients under standard care                    |
| Sample     | Pilot n≈50–500 (pseudonymous EP001...)                                       |
| Data       | Scalp EEG (EDF), role assessments, severity labels — de■identified           |
| Duration   | 12 months + retention per policy                                             |
| Risk level | Minimal (no change to clinical care; de■identified analysis)                 |

## 2. Background & aims

Seizure detection/severity support using real EEG (e.g., CHB■MIT■style scalp EEG) + structured clinician assessments, with explainable (SHAP/permutation) outputs and human■in■the■loop review.

Aim 1: build a subject■level seizure■detection model. Aim 2: fuse EEG + assessment for severity.

Aim 3: evaluate fairness, explainability, and clinical utility.

## 3. Procedures

1. Eligible patients receive the **consent form + EULA** (§Consent).

2. On e■signed consent, data is collected under routine care (no extra procedures).
3. Data is **de■identified (HIPAA Safe■Harbor)** to a pseudonym before analysis.
4. Models are trained/evaluated on de■identified data with subject■level splits.
5. A neurophysiologist reviews all AI outputs; **no autonomous diagnosis**.

## 4. Risks, benefits, minimisation

| Risk                         | Likelihood | Minimisation                                                  |
|------------------------------|------------|---------------------------------------------------------------|
| Privacy/PHI breach           | Low        | encryption, de■identification, RBAC, audit (see Security doc) |
| Re■identification            | Low        | Safe■Harbor + no linkage keys in analysis env                 |
| Model error influencing care | Low        | decision■support only + mandatory clinician review            |
| Group harm / bias            | Low        | fairness evaluation (subgroup metrics) + reporting            |

**Benefits:** potential improved seizure■risk stratification; societal knowledge. No direct guarantee to participant.

## 5. Informed consent process

- Written + electronic consent; capacity assessed; assent + parental permission for minors.
- Voluntary; withdrawal any time without affecting care.
- Consent covers **specific use**: model development, validation, and de■identified secondary analysis.
- Documented in the platform with timestamp, version, and revocation status.

## 6. Data management & confidentiality

- Storage: encrypted (AES■256), access■controlled, audited.
- Sharing: de■identified only; no PHI leaves the secure zone.
- Retention: per policy; secure destruction at end of retention.
- Governance: Responsible■AI pillars + Continuous■Monitoring.

## 7. Equitable selection

Inclusion regardless of sex, race, socioeconomic status; paediatric inclusion justified (epilepsy onset is often paediatric) with added protections.

## Diagrams

### Flowchart — IRB + consent gate before any data use

■ Diagram (Mermaid — interactive in the viewer): flowchart TD

### Sequence — consent-to-analysis

■ Diagram (Mermaid — interactive in the viewer): sequenceDiagram

### Network — oversight relationships

■ Diagram (Mermaid — interactive in the viewer): graph LR

## ***Journey — participant experience***

■ Diagram (Mermaid — interactive in the viewer): journey

**Reason:** obtain ethical approval. **Why:** human■subjects research is regulated (45 CFR 46 / Common Rule).

**What is happening:** minimal■risk study with valid consent + de■identification. **How it is happening:**

consent gate → de■identify → subject■level modelling → clinician■reviewed outputs. **Reference:** HHS (2018, Common Rule); WMA (2013, Declaration of Helsinki).

## **Professor Readiness (Defense Q&A)**

### ***Why is this minimal risk?***

No change to clinical care; analysis is on de■identified data; outputs are clinician■reviewed decision■support.

### ***How do minors get protected?***

Parental permission + assent + additional safeguards (45 CFR 46 Subpart D).

### ***What is the withdrawal effect?***

Withdrawal any time, no impact on care; already■de■identified aggregated data may be retained per protocol.

## **References**

HHS. (2018). *Federal Policy for the Protection of Human Subjects (Common Rule)*, 45 CFR 46.

World Medical Association. (2013). *Declaration of Helsinki: Ethical principles for medical research involving human subjects*. *JAMA*, 310(20), 2191–2194.



# Patient Consent + EULA — Data Use for Epilepsy AI Analysis

**Why (this doc):** Before any patient's epilepsy data (EEG, assessments) is used for AI analysis, the platform must capture **informed consent** and agreement to an **End-User Licence Agreement (EULA) governing data use. This is the consent/EULA text and the capture workflow.** How: the viewer's Survey / AI Forms tabs present this as a gated first step; consent is versioned, timestamped, and revocable, per the IRB pack and Security doc.

## 1. Informed consent (plain language)

*Caption — the core disclosures a participant must understand before agreeing.*

| Element            | Statement                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------|
| Purpose            | Your EEG and assessment answers may be used to develop and test AI that supports epilepsy care. |
| Voluntary          | Participation is voluntary. Declining will <b>not</b> affect your care.                         |
| What we collect    | EEG recordings, questionnaire answers, and severity information.                                |
| De-identification  | Your identity is removed and replaced with a code (e.g., EP001) before analysis.                |
| Who sees it        | Your care team and authorised researchers under strict security.                                |
| AI is support only | AI never diagnoses you. A neurologist reviews all results.                                      |
| Risks              | Main risk is privacy; we encrypt and de-identify to minimise it.                                |
| Benefits           | You may not directly benefit; findings may help future patients.                                |
| Withdrawal         | You may withdraw any time; we stop using new data and honour deletion where feasible.           |
| Contact            | PI / IRB contact details for questions or complaints.                                           |

**Consent statement:** "I have read and understood the above. I voluntarily consent to the collection, de-identification, and use of my epilepsy data for the described AI research." — e-signature + date + version.

## 2. EULA (data-use licence) — key clauses

*Caption — the licence terms governing how the platform may use the de-identified data.*

| Clause           | Term                                                                                                              |
|------------------|-------------------------------------------------------------------------------------------------------------------|
| Licence grant    | Non-exclusive licence to use <b>de-identified</b> data for model development, validation, and secondary analysis. |
| Scope limitation | No re-identification attempts; no sale of data; no use outside the stated research purpose.                       |
| Security         | Data protected per HIPAA/NIST/OWASP controls (encryption, RBAC, audit).                                           |

| Clause                 | Term                                                                                   |
|------------------------|----------------------------------------------------------------------------------------|
| Retention              | Data retained only for the retention period, then securely destroyed.                  |
| Revocation             | Consent may be revoked; downstream use of new data ceases.                             |
| Liability & disclaimer | AI output is decision support, not medical advice; no warranty of diagnostic accuracy. |
| Governing terms        | Subject to IRB oversight and applicable law (HIPAA, Common Rule).                      |
| Acceptance             | Explicit "I Agree" required; agreement is logged (who/when/version).                   |

### 3. Consent capture workflow

1. Present consent + EULA (this doc) in the patient's language, readable format.
2. Verify comprehension (teach back / questions).
3. Capture **e-signature** + timestamp + document version.
4. Store consent record (immutable, audited); set participant status = *consented*.
5. Only *consented* participants' data flows to deidentification → analysis.
6. Revocation flips status; pipeline excludes new data.

## Diagrams

### Flowchart — consent gate

■ Diagram (Mermaid — interactive in the viewer): flowchart TD

### Sequence — e-consent

■ Diagram (Mermaid — interactive in the viewer): sequenceDiagram

### Network — consent record relationships

■ Diagram (Mermaid — interactive in the viewer): graph LR

### Journey — consent experience

■ Diagram (Mermaid — interactive in the viewer): journey

**Reason:** obtain lawful, ethical basis to use patient data. **Why:** without consent + EULA, data use is unlawful and unethical. **What is happening:** patient is informed, agrees, and can revoke; only consented, deidentified data is analysed. **How it is happening:** gated e-consent → immutable log → pipeline gate. **Reference:** HHS (2018); WMA (2013); GDPR Art. 7 (where applicable).

## Professor Readiness (Defense Q&A)

### Is EULA acceptance enough?

No — EULA governs data use licence; **informed consent** (IRB approved) is the ethical basis. Both are captured.

### ***How is revocation honoured technically?***

Consent status is a pipeline gate; on revocation, new data is excluded and deletion is honoured where feasible (aggregates may persist).

### ***Where does the UI enforce this?***

The *Survey / AI Forms* tabs present consent as step 0; analysis datasets carry only consented pseudonyms.

## **References**

HHS. (2018). *Federal Policy for the Protection of Human Subjects (Common Rule)*, 45 CFR 46.

World Medical Association. (2013). *Declaration of Helsinki*. *JAMA*, 310(20), 2191–2194.

European Parliament. (2016). *General Data Protection Regulation (EU 2016/679)*, Art. 7.

# Model Lifecycle — Phase-Gate Scorecard (Quality · Scoring · Monitoring)

**Why (this doc):** Each phase of the model lifecycle is quality-checked against real repo artefacts, scored 0-100, RAG-rated, and assigned a monitoring signal — a per-phase governance dashboard. **How:** `mlops/phase_gates.py` runs evidence-based checks; the scorecard CSV also renders in the viewer Data tab.

**Overall maturity: 100/100**

Weakest phase: **1. Problem definition** (100/100) — see gaps below.

## Phase scorecard — pipeline · monitoring · quality · score · status · visualization

| phase                          | pipeline                           | monitoring               | quality_checks | score | status | visualization                           |
|--------------------------------|------------------------------------|--------------------------|----------------|-------|--------|-----------------------------------------|
| 1. Problem definition          | docs/research-problems.md          | n/a                      | 3/3            | 100   | Green  | docs/research-framework.md              |
| 2. Data ingestion              | analysis/make_cohort.py            | freshness / record count | 3/3            | 100   | Green  | viewer Data tab                         |
| 3. Data validation             | mlops/data_quality.py              | contract violations      | 3/3            | 100   | Green  | docs/analysis/data-quality-report.md    |
| 4. Data preparation            | analysis/preprocessing.py          | null% / outlier%         | 4/4            | 100   | Green  | docs/analysis/eda-report.md             |
| 5. Feature engineering / store | mlops/feature_store.py             | feature drift            | 3/3            | 100   | Green  | docs/analysis/variable-dictionary.md    |
| 6. Model development           | analysis/primary_analysis.py       | n/a                      | 4/4            | 100   | Green  | docs/analysis/primary-analysis.md       |
| 7. Training                    | analysis/run_all.py                | n/a                      | 3/3            | 100   | Green  | docs/analysis/primary-analysis.md       |
| 8. Evaluation / validation     | analysis/eda.py                    | performance decay        | 6/6            | 100   | Green  | docs/analysis/eda-report.md (charts)    |
| 9. Explainability / fairness   | analysis/responsible_ai_runtime.py | fairness gap             | 3/3            | 100   | Green  | docs/analysis/responsible-ai-runtime.md |

| phase                                | pipeline                     | monitoring           | quality_checks | score | status | visualization                               |
|--------------------------------------|------------------------------|----------------------|----------------|-------|--------|---------------------------------------------|
| 10. Deployment / serving             | api/main.py                  | latency / error rate | 4/4            | 100   | Green  | GET /metrics                                |
| 11. Monitoring / observability       | mlops/observability.py       | drift / system / API | 4/4            | 100   | Green  | docs/analysis/observability-report.md       |
| 12. Governance                       | docs/responsible-ai/index.md | audit / compliance   | 4/4            | 100   | Green  | docs/responsible-ai/implementation/index.md |
| 13. Retraining / continuous learning | mlops/retrain.py (planned)   | champion-challenger  | 3/3            | 100   | Green  | docs/phase-gates-scorecard.md               |

\_Dashboard: this scorecard + data/analysis/phase\_scorecard.csv (renders in the viewer **Data** tab).\_

## Statistical summary

- Phases: 13 · mean score 100 · median 100 · min 100 · max 100
- Green 13 · Amber 0 · Red 0

## Per-phase quality checks (PASS / GAP)

| phase                          | quality_check                           | result |
|--------------------------------|-----------------------------------------|--------|
| 1. Problem definition          | research problems documented            | PASS   |
| 1. Problem definition          | hypotheses (IV->DV->test)               | PASS   |
| 1. Problem definition          | variable dictionary (DV/IV/covariate)   | PASS   |
| 2. Data ingestion              | cohort materialised                     | PASS   |
| 2. Data ingestion              | data contract defined                   | PASS   |
| 2. Data ingestion              | REAL EEG dataset fetched                | PASS   |
| 3. Data validation             | data-quality catalogue                  | PASS   |
| 3. Data validation             | quality report                          | PASS   |
| 3. Data validation             | contract enforced                       | PASS   |
| 4. Data preparation            | preprocessing library                   | PASS   |
| 4. Data preparation            | cleaned features persisted              | PASS   |
| 4. Data preparation            | imputation + outliers + encoding        | PASS   |
| 4. Data preparation            | fitted preprocessing pipeline persisted | PASS   |
| 5. Feature engineering / store | feature store                           | PASS   |
| 5. Feature engineering / store | feature metadata catalogue              | PASS   |
| 5. Feature engineering / store | derived features                        | PASS   |

| phase                                | quality_check                      | result |
|--------------------------------------|------------------------------------|--------|
| 6. Model development                 | classification model               | PASS   |
| 6. Model development                 | survival model                     | PASS   |
| 6. Model development                 | time-series model                  | PASS   |
| 6. Model development                 | hyperparameter optimization        | PASS   |
| 7. Training                          | subject-level split                | PASS   |
| 7. Training                          | cross-validation                   | PASS   |
| 7. Training                          | experiment tracking                | PASS   |
| 8. Evaluation / validation           | AUC / discrimination               | PASS   |
| 8. Evaluation / validation           | calibration (Brier)                | PASS   |
| 8. Evaluation / validation           | survival C-index                   | PASS   |
| 8. Evaluation / validation           | decision curve analysis            | PASS   |
| 8. Evaluation / validation           | bootstrap CIs + DeLong + nested CV | PASS   |
| 8. Evaluation / validation           | external validation (REAL data)    | PASS   |
| 9. Explainability / fairness         | SHAP + LIME                        | PASS   |
| 9. Explainability / fairness         | fairness + mitigation              | PASS   |
| 9. Explainability / fairness         | responsible-AI pillars             | PASS   |
| 10. Deployment / serving             | API service                        | PASS   |
| 10. Deployment / serving             | containerised + healthcheck        | PASS   |
| 10. Deployment / serving             | model registry + rollback          | PASS   |
| 10. Deployment / serving             | /predict serves registered model   | PASS   |
| 11. Monitoring / observability       | data/model observability           | PASS   |
| 11. Monitoring / observability       | system monitor (CPU/mem/GPU)       | PASS   |
| 11. Monitoring / observability       | API metrics endpoint               | PASS   |
| 11. Monitoring / observability       | logging (app/inference/audit)      | PASS   |
| 12. Governance                       | responsible-AI governance          | PASS   |
| 12. Governance                       | audit logging                      | PASS   |
| 12. Governance                       | prompt log / provenance            | PASS   |
| 12. Governance                       | auto-generated model cards         | PASS   |
| 13. Retraining / continuous learning | retraining pipeline                | PASS   |
| 13. Retraining / continuous learning | champion-challenger                | PASS   |
| 13. Retraining / continuous learning | feedback capture                   | PASS   |

**Open gaps (GAP rows)** are the honest to-do list per phase: fitted-pipeline persistence, hyperparameter optimization, decision-curve analysis, external validation, /predict model serving, auto-generated model cards, and the retraining / champion-challenger loop.